
An Efficient Algorithm for Thresholding Monte Carlo Tree Search

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Abstract

1 We introduce the Thresholding Monte Carlo Tree Search problem, in which, given
2 a tree $\mathcal{T}$ and a threshold θ , a player must answer whether the root node value
3 of $\mathcal{T}$ is at least θ or not. In the given tree, ‘MAX’ or ‘MIN’ is labeled on each
4 internal node, and the value of a ‘MAX’-labeled (‘MIN’-labeled) internal node is
5 the maximum (minimum) of its child values. The value of a leaf node is the mean
6 reward of an unknown distribution, from which the player can sample rewards. For
7 this problem, we develop a δ -correct sequential sampling algorithm based on the
8 Track-and-Stop strategy that has asymptotically optimal sample complexity. We
9 show that a ratio-based modification of the D-Tracking arm-pulling strategy leads
10 to a substantial improvement in empirical sample complexity, as well as reducing
11 the per-round computational cost from linear to logarithmic in the number of arms.

12 1 Introduction

13 The last two decades saw the success of Monte Carlo Tree Search (MCTS) in many domains,
14 including abstract games [Coulom, 2006, Silver et al., 2016, Yoshizoe and Müller, 2019], planning
15 and exploration of Markov decision process (MDP) structure [Chang et al., 2005, Pepels et al., 2014,
16 Best et al., 2018, Leurent and Maillard, 2020, Jonsson et al., 2020], solving NP-hard problems Xing
17 and Tu [2020], Yang et al. [2021], Khalil et al. [2022], Abe et al. [2022], as well as guiding through
18 large language models [Yao et al., 2023, Wan et al., 2024]. In particular, the most widely used method
19 in MCTS incorporates a bandit-based method Kocsis and Szepesvári [2006]. Bandit-based methods
20 framed MCTS as a solution to the exploration and exploitation tradeoff. However, in many problems,
21 what we truly seek is the most effective action in the tree, and in this sense, MCTS can be viewed as
22 a pure exploration problem Tolpin and Shimony [2012]. As is well known, bandit-based methods
23 are suboptimal for pure exploration problems Bubeck et al. [2009], as they lack the robustness in
24 identifying the best action.

25 A key difficulty in understanding the theoretical property of purely exploring MCTS is that it
26 intertwines two aspects: tree expansion (constructing a search tree adaptively) and exploration on
27 the currently expanded tree (allocating samples to evaluate candidate actions/states). Analyzing
28 algorithms that dynamically expand the tree is theoretically appealing but challenging, as it typically
29 requires nontrivial assumptions to relate statistics at an internal node to those of its descendants
30 [Coquelin and Munos, 2007]. As a result, much of the recent theoretical literature separates these
31 two aspects and focuses on pure exploration of a given (expanded) tree, treating the tree structure
32 as fixed and using this as a fundamental modeling assumption [Tolpin and Shimony, 2012, Teraoka
33 et al., 2014, Feldman and Domshlak, 2014, Kaufmann and Koolen, 2017].

34 Most practical tree structures are astronomically large. For example, the entire tree size of the
35 Game of Go is estimated to be around 10^{172} , making it impossible to evaluate all nodes exhaustively
36 Yoshizoe and Müller [2019]. Instead, we explicitly construct the search tree only up to a fixed depth

or according to a predefined expansion criterion. From the resulting frontier nodes (leaf nodes of the constructed tree), the remaining part of the tree is evaluated via stochastic playouts, i.e., randomized rollouts that simulate the game until termination. From this viewpoint, each leaf node corresponds not to a single deterministic value, but to an unknown reward distribution.

Seminal work by Kaufmann and Koolen [2017] considers an identification of ε -best action in this setting with a generalization of the δ -correct best arm identification [Bechhofer et al., 1968, Glynn and Juneja, 2004, Garivier and Kaufmann, 2016]. The work derived an algorithm based on the confidence bound strategy and proves its bound on the sample complexity (stopping time). However, the optimal sample complexity they derived does not admit a closed-form expression, nor does their algorithm have asymptotic optimality.

Similar to Kaufmann and Koolen [2017], we consider the fixed-confidence setting, where the objective is to design a δ -correct algorithm that minimizes the sample complexity. We depart from their setting by introducing a threshold θ and aim to decide whether the root value exceeds θ with probability at least $1 - \delta$, following the thresholding perspective studied in prior work [Locatelli et al., 2016, Kano et al., 2019, Tabata et al., 2023, Tsai et al., 2025]. Thanks to this threshold introduction, sample complexity is lower bounded by the value that is calculated by a simple recursive formula (Section 3). As a result, the optimal sampling proportion of arms can be calculated in time linear to the tree depth, which enables to design an efficient algorithm that tracks the proportion.

Our algorithm design starts from the Track-and-Stop algorithm Garivier and Kaufmann [2016], Degenne and Koolen [2019] that uses the optimal sample proportion estimated by plug-in mean parameters. However, the standard Track-and-Stop algorithm has two major shortcomings. First, it is sensitive to the estimation error of the plug-in estimators that degrades the empirical sample complexity. Second, it requires recomputing the optimal sampling proportions at every step, which requires $O(|\mathcal{L}(\mathcal{T})|)$ time in the number of leaves $|\mathcal{L}(\mathcal{T})|$ in the tree $\mathcal{T}$. To address these, we introduce **the ratio-based draw**, a simple modification to the Track-and-Stop algorithm. This modification to D-Tracking reduces the empirical sample complexity by several factors (Section 6). Moreover, the ratio-based draw, combined with careful construction of the data structure, enables the modified D-Tracking to run in $O(\log |\mathcal{L}(\mathcal{T})|)$ time per step if $\mathcal{T}$ is balanced, without resorting to approximate computations (Section 5.4).

We deal with the decision problem but the typical setting on MCTS is the identification problem, that is, the problem of the best next action identification. We also show that our slightly modified version of our algorithm is an optimal algorithm for the identification of a good next action.

2 Problem Setting

We study a binary decision problem with outcomes in $\{\text{'win'}, \text{'lose'}\}$ defined on a rooted tree $\mathcal{T} = (S, B, s_0)$, where S is the set of nodes, B is the set of branches, and s_0 is the root node. Each node represents a state, and s_0 is the current state. A branch (s_1, s_2) corresponds to an action at state s_1 that leads to state s_2 . A node is a leaf if it has no children. A node is internal if it is not a leaf. Each internal node s has label $L(s) \in \{\text{'MAX'}, \text{'MIN'}\}$.

Let $\mathcal{L}(\mathcal{T})$ be the set of leaf nodes in $\mathcal{T}$. Each leaf corresponds to an arm, and we use the terms *leaf* and *arm* interchangeably. Given a question tree $\mathcal{T}$ and its internal node label function L , at each round $t = 1, 2, \dots$, the player selects one leaf node $I(t) \in \mathcal{L}(\mathcal{T})$, and observes a reward $r(t) \in X \subseteq \mathbb{R}$ drawn from an unknown distribution with mean $\mu_{I(t)} \in X$. In this paper, we assume that the distributions belong to a one-parameter exponential family, and the rewards observed from the same leaf are i.i.d. For each $\ell \in \mathcal{L}(\mathcal{T})$, $\hat{\mu}_\ell(t)$ and $N_\ell(t) = |\{s \in [1, t] : I(s) = \ell\}|$ denote the sample mean of the rewards and the number of selections, respectively, for the leaf node ℓ by round t .

Throughout the paper, we denote $(\mu_\ell)_{\ell \in \mathcal{L}(\mathcal{T})} \in X^{|\mathcal{L}(\mathcal{T})|}$ by $\boldsymbol{\mu}$. For any vector $\boldsymbol{\lambda} = (\lambda_\ell)_{\ell \in \mathcal{L}(\mathcal{T})} \in X^{|\mathcal{L}(\mathcal{T})|}$, we denote $\boldsymbol{\lambda}_{\mathcal{L}' } = (\lambda_\ell)_{\ell \in \mathcal{L}' }$ for any subset of leaves $\mathcal{L}' \subseteq \mathcal{L}(\mathcal{T})$. Let $\mathcal{D}(s) \subseteq \mathcal{L}(\mathcal{T})$ be the set of leaf nodes that are descendants of node s . For each node $s \in S$, we define the *value* $V_s(\boldsymbol{\mu})$ of node s based on $\boldsymbol{\mu}_{\mathcal{D}(s)}$ as

$$V_s(\boldsymbol{\mu}) = \begin{cases} \mu_s & (s \in \mathcal{L}(\mathcal{T})), \\ \max_{c \in \mathcal{C}(s)} V_c(\boldsymbol{\mu}) & (L(s) = \text{'MAX'}), \\ \min_{c \in \mathcal{C}(s)} V_c(\boldsymbol{\mu}) & (L(s) = \text{'MIN'}), \end{cases}$$

86 where $\mathcal{C}(s)$ is the set of children of s .

87 Whether the answer for the question tree $\mathcal{T}$ is ‘win’ or ‘lose’ depends on a given threshold $\theta \in X$.
 88 We call the nodes $s \in S$ with $V_s(\boldsymbol{\mu}) \geq \theta$ good (winning) nodes, and the other nodes bad (losing)
 89 nodes. The player’s task is to determine whether the root node s_0 is a winning or losing node. Let
 90 $A_s(\boldsymbol{\mu}) = \{c \in \mathcal{C}(s) \mid V_c(\boldsymbol{\mu}) \geq \theta\}$ be the set of good child nodes for each node $s \in S$. Here, we
 91 define $a_s(\boldsymbol{\mu})$ by

$$a_s(\boldsymbol{\mu}) = \begin{cases} \text{‘win’} & (L(s) = \text{‘MAX’}, A_s(\boldsymbol{\mu}) \neq \emptyset \text{ or } L(s) = \text{‘MIN’}, A_s(\boldsymbol{\mu}) = \mathcal{C}(s)), \\ \text{‘lose’} & (L(s) = \text{‘MAX’}, A_s(\boldsymbol{\mu}) = \emptyset \text{ or } L(s) = \text{‘MIN’}, A_s(\boldsymbol{\mu}) \neq \mathcal{C}(s)), \end{cases}$$

92 and $a_s(\boldsymbol{\mu}) = \text{‘win’}$ is equivalent to $V_s(\boldsymbol{\mu}) \geq \theta$.

93 The function values $V_s(\boldsymbol{\mu})$, $A_s(\boldsymbol{\mu})$ and $a_s(\boldsymbol{\mu})$ depend solely on $\boldsymbol{\mu}_{\mathcal{D}(s)}$. Thus, we abuse the notations
 94 and use the function values even for $\boldsymbol{\lambda} = (\lambda_\ell)_{\ell \in \mathcal{D}(s)} \in X^{|\mathcal{D}(s)|}$ as $V_s(\boldsymbol{\lambda})$, $A_s(\boldsymbol{\lambda})$, and $a_s(\boldsymbol{\lambda})$.

95 At the end of any round, the player can choose to stop and output the estimated answer ‘win’ or
 96 ‘lose.’ The player should follow an algorithm to decide which arm to select as well as when to stop.
 97 Since the rewards are generated stochastically, it is impossible to output a correct answer with 100%
 98 accuracy. In the fixed confidence setting, the objective is relaxed to δ -correctness, defined as follows.

99 **Definition 2.1** (δ -correctness). Given a rooted tree $\mathcal{T} = (S, B, s_0)$, its internal node label function
 100 L , and $\delta > 0$, an algorithm is said to be δ -correct if the algorithm outputs the correct answer $a_{s_0}(\boldsymbol{\mu})$
 101 with probability at least $1 - \delta$ by sampling rewards from the leaves for any unknown mean vector $\boldsymbol{\mu}$
 102 of the leaf reward distributions.

103 Let τ_δ denote the number of reward samples (stopping time) for a given confidence parameter $\delta > 0$.
 104 The problem that we consider in this paper is formally stated as follows.

105 **Problem 2.2.** Design a δ -correct algorithm with minimum $\mathbb{E}[\tau_\delta]$ for a given rooted tree $\mathcal{T} =$
 106 (S, B, s_0) , its internal node label function L , and $\delta > 0$.

107 3 Optimal Allocation

108 In this section, we describe the optimal sampling proportions that achieve the asymptotically optimal
 109 stopping time with fixed confidence.

110 We define the alternative set, i.e., the set of mean vectors of leaf rewards that induces a different
 111 answer than $\boldsymbol{\mu}$ as

$$\text{Alt}(\boldsymbol{\mu}) = \{\boldsymbol{\lambda} = (\lambda_\ell)_{\ell \in \mathcal{L}(\mathcal{T})} \in X^{|\mathcal{L}(\mathcal{T})|} \mid a_{s_0}(\boldsymbol{\lambda}) \neq a_{s_0}(\boldsymbol{\mu})\}.$$

112 Let $d(x, y)$ be the Kullback-Leibler divergence between two distributions (in the same exponential
 113 family) with mean x and y . Following Theorem 1 of Garivier and Kaufmann [2016], for any δ -correct
 114 algorithm, we can construct the asymptotic lower bound of the expected stopping time $\mathbb{E}[\tau_\delta]$.

115 **Theorem 3.1** (Garivier and Kaufmann [2016], Degenne and Koolen [2019]). *The following inequality*
 116 *holds:*

$$\liminf_{\delta \rightarrow 0} \frac{\mathbb{E}_{\boldsymbol{\mu}}[\tau_\delta]}{\log(1/\delta)} \geq \left(\max_{\boldsymbol{w} \in \Delta} \inf_{\boldsymbol{\lambda} \in \text{Alt}(\boldsymbol{\mu})} \sum_{\ell \in \mathcal{L}(\mathcal{T})} w_\ell d(\mu_\ell, \lambda_\ell) \right)^{-1}, \quad (1)$$

117 where Δ is the simplex over leaves $\Delta = \{(w_\ell)_{\ell \in \mathcal{L}(\mathcal{T})} \mid \sum_\ell w_\ell = 1, w_\ell \geq 0 (\ell \in \mathcal{L}(\mathcal{T}))\}$.

118 Ineq. (1) involves an optimization problem over $\boldsymbol{w} \in \Delta$, and the maximizer

$$\boldsymbol{w}(\boldsymbol{\mu}) = (w_\ell(\boldsymbol{\mu}))_{\ell \in \mathcal{L}(\mathcal{T})} = \arg \max_{\boldsymbol{w} \in \Delta} \inf_{\boldsymbol{\lambda} \in \text{Alt}(\boldsymbol{\mu})} \sum_{\ell \in \mathcal{L}(\mathcal{T})} w_\ell d(\mu_\ell, \lambda_\ell) \quad (2)$$

119 is the asymptotically optimal sampling proportion over leaves.

120 To solve the optimization problem of (2) recursively, we consider a subproblem for the subtree rooted
 121 at node s . Define the restrictions of $\text{Alt}(\boldsymbol{\mu})$ and Δ to the descendant leaves $\mathcal{D}(s)$, denoted by $\text{Alt}_s(\boldsymbol{\mu})$
 122 and Δ_s , respectively, as

$$\text{Alt}_s(\boldsymbol{\mu}) = \{\boldsymbol{\lambda} = (\lambda_\ell)_{\ell \in \mathcal{D}(s)} \in X^{|\mathcal{D}(s)|} \mid a_s(\boldsymbol{\lambda}) \neq a_s(\boldsymbol{\mu})\}$$

Recursive Formula 1 Calculation of $\tilde{d}_s(\mu)$, $w^s(\mu)$ and $\tilde{Z}_s(t)$

$$\begin{aligned}
 \tilde{d}_s(\mu) &= \begin{cases} d(\mu_s, \theta) & (s \in \mathcal{L}(T), \mu_s \geq \theta) \\ -d(\mu_s, \theta) & (s \in \mathcal{L}(T), \mu_s < \theta) \\ \max_{c \in \mathcal{C}(s)} \tilde{d}_c(\mu) & \begin{pmatrix} L(s) = \text{'MAX'}, \\ a_s(\mu) = \text{'win'} \end{pmatrix} \\ \frac{1}{\sum_{c \in \mathcal{C}(s)} 1/\tilde{d}_c(\mu)} & \begin{pmatrix} L(s) = \text{'MAX'}, \\ a_s(\mu) = \text{'lose'} \end{pmatrix} \\ \frac{1}{\sum_{c \in \mathcal{C}(s)} 1/\tilde{d}_c(\mu)} & \begin{pmatrix} L(s) = \text{'MIN'}, \\ a_s(\mu) = \text{'win'} \end{pmatrix} \\ \min_{c \in \mathcal{C}(s)} \tilde{d}_c(\mu) & \begin{pmatrix} L(s) = \text{'MIN'}, \\ a_s(\mu) = \text{'lose'} \end{pmatrix} \end{cases} \quad (5) \\
 \tilde{Z}_s(t) &= \begin{cases} N_s(t)d(\hat{\mu}_s(t), \theta) & (s \in \mathcal{L}(T), \hat{\mu}_s(t) \geq \theta) \\ -N_s(t)d(\hat{\mu}_s(t), \theta) & (s \in \mathcal{L}(T), \hat{\mu}_s(t) < \theta) \\ \sum_{c \in \mathcal{C}(s), \tilde{Z}_c(t) > 0} \tilde{Z}_c(t) & \begin{pmatrix} L(s) = \text{'MAX'}, \\ a_s(\hat{\mu}(t)) = \text{'win'} \end{pmatrix} \\ \max_{c \in \mathcal{C}(s)} \tilde{Z}_c(t) & \begin{pmatrix} L(s) = \text{'MAX'}, \\ a_s(\hat{\mu}(t)) = \text{'lose'} \end{pmatrix} \\ \min_{c \in \mathcal{C}(s)} \tilde{Z}_c(t) & \begin{pmatrix} L(s) = \text{'MIN'}, \\ a_s(\hat{\mu}(t)) = \text{'win'} \end{pmatrix} \\ \sum_{c \in \mathcal{C}(s), \tilde{Z}_c(t) < 0} \tilde{Z}_c(t) & \begin{pmatrix} L(s) = \text{'MIN'}, \\ a_s(\hat{\mu}(t)) = \text{'lose'} \end{pmatrix} \end{cases} \\
 w_\ell^s(\mu) &= \begin{cases} 1 & (s \in \mathcal{L}(T)) \\ w_\ell^{c^+(s)}(\mu) & \begin{pmatrix} L(s) = \text{'MAX'}, \\ a_s(\mu) = \text{'win'}, \\ \ell \in \mathcal{D}(c^+(s)) \end{pmatrix} \\ 0 & \begin{pmatrix} L(s) = \text{'MAX'}, \\ a_s(\mu) = \text{'win'}, \\ \ell \in \mathcal{D}(c), \\ c \in \mathcal{C}(s) \setminus \{c^+(s)\} \end{pmatrix} \\ \frac{w_\ell^c(\mu)/\tilde{d}_c(\mu)}{\sum_{c' \in \mathcal{C}(s)} 1/\tilde{d}_{c'}(\mu)} & \begin{pmatrix} L(s) = \text{'MAX'}, \\ a_s(\mu) = \text{'lose'}, \\ \ell \in \mathcal{D}(c), c \in \mathcal{C}(s) \end{pmatrix} \\ \frac{w_\ell^c(\mu)/\tilde{d}_c(\mu)}{\sum_{c' \in \mathcal{C}(s)} 1/\tilde{d}_{c'}(\mu)} & \begin{pmatrix} L(s) = \text{'MIN'}, \\ a_s(\mu) = \text{'win'}, \\ \ell \in \mathcal{D}(c), c \in \mathcal{C}(s) \end{pmatrix} \\ w_\ell^{c^-(s)}(\mu) & \begin{pmatrix} L(s) = \text{'MIN'}, \\ a_s(\mu) = \text{'lose'}, \\ \ell \in \mathcal{D}(c^-(s)) \end{pmatrix} \\ 0 & \begin{pmatrix} L(s) = \text{'MIN'}, \\ a_s(\mu) = \text{'lose'}, \\ \ell \in \mathcal{D}(c), \\ c \in \mathcal{C}(s) \setminus \{c^-(s)\} \end{pmatrix} \end{cases} \\
 &\quad \text{where } c^+(s) = \arg \max_{c \in \mathcal{C}(s)} \tilde{d}_c(\mu) \text{ and} \\
 &\quad c^-(s) = \arg \min_{c \in \mathcal{C}(s)} \tilde{d}_c(\mu)
 \end{aligned}$$

123 and $\Delta_s = \left\{ (w_\ell)_{\ell \in \mathcal{D}(s)} \mid \sum_{\ell \in \mathcal{D}(s)} w_\ell = 1, w_\ell \geq 0 \ (\ell \in \mathcal{D}(s)) \right\}$. We consider the following opti-
 124 mization problem for the subtree rooted at s .

$$w^s(\mu) = \arg \max_{w \in \Delta_s} \inf_{\lambda \in \text{Alt}_s(\mu)} \sum_{\ell \in \mathcal{D}(s)} w_\ell d(\mu_\ell, \lambda_\ell). \quad (3)$$

125 Let $d_s(\mu)$ be the optimal value of the optimization problem of (3), that is,

$$d_s(\mu) = \max_{w \in \Delta_s} \inf_{\lambda \in \text{Alt}_s(\mu)} \sum_{\ell \in \mathcal{D}(s)} w_\ell d(\mu_\ell, \lambda_\ell). \quad (4)$$

126 **Theorem 3.2.** The vector $w^s(\mu)$ and the scalar $d_s(\mu) = |\tilde{d}_s(\mu)|$ calculated by Recursive Formula 1
 127 are the solutions of the optimization problem defined by Eqs. (3) and (4). Furthermore,

$$\tilde{d}_s(\mu) \geq 0 \Leftrightarrow a_s(\mu) = \text{'win'} \quad (\tilde{d}_s(\mu) < 0 \Leftrightarrow a_s(\mu) = \text{'lose'}).$$

128 The proof of this Theorem is provided in Appendix A.

129 Therefore, Ineq. (1) is rewritten as

$$\liminf_{\delta \rightarrow 0} \frac{\mathbb{E}[\tau_\delta]}{\log(1/\delta)} \geq \frac{1}{d_{s_0}(\mu)}.$$

130 4 Track-and-Stop Algorithms

131 This section introduces RD-Tracking-TMCTS, a Track-and-Stop algorithm for the thresholding
 132 MCTS problem. A Track-and-Stop algorithm Garivier and Kaufmann [2016], Degenne and Koolen

[2019] draws arms proportional to the optimal weights (Eq. (2)) where the true mean vector is replaced by its empirical analogue. The design of Track-and-Stop algorithms is done by determining a sampling rule, a stopping rule, and a recommendation rule. The recommendation rule we adopt here is very simple. We recommend $a_{s_0}(\hat{\mu}(t))$ as the answer for the given $\mathcal{T}$. We next describe our design of stopping and sampling rules.

4.1 Stopping Rule

To guarantee δ -correctness, we adopt the GLR (Generalized Likelihood Ratio) stopping rule [Kaufmann and Koolen, 2021].

Let $R(t) = \{r(u)\}_{u=1,\dots,t}$ be the set of observed rewards until time t , and $R_s(t) = \{r(u) \in R(t) \mid I(u) \in \mathcal{D}(s)\}$ be the set of rewards observed in the descendants of node s .

To evaluate the reliability of $a_s(\hat{\mu}_{\mathcal{D}(s)}(t))$, we introduce the Generalized Likelihood Ratio (GLR) statistic $Z_s(t)$:

$$Z_s(t) = \log \frac{\sup_{\lambda \in X^{\mathcal{D}(s)}} \Lambda(\lambda \mid R_s(t))}{\sup_{\lambda \in \text{Alt}_s(\hat{\mu}(t))} \Lambda(\lambda \mid R_s(t))} = \inf_{\lambda \in \text{Alt}_s(\hat{\mu}(t))} \sum_{\ell \in \mathcal{D}(s)} N_\ell(t) d(\hat{\mu}_\ell(t), \lambda_\ell) \quad (6)$$

where $\Lambda(\lambda \mid R_s(t))$ is the likelihood of λ given the set of observed rewards $R_s(t)$.

GLR statistic $Z_s(t)$ for any node s can be computed recursively using Recursive Formula 1.

Theorem 4.1. *For any node $s \in S$, the GLR statistic $Z_s(t)$ defined by Eq. (6) is equal to the absolute value of $\tilde{Z}_s(t)$, which is computed using Recursive Formula 1. Furthermore,*

$$\tilde{Z}_s(t) \geq 0 \Leftrightarrow a_s(\hat{\mu}(t)) = \text{'win'} \quad (\tilde{Z}_s(t) < 0 \Leftrightarrow a_s(\hat{\mu}(t)) = \text{'lose'}).$$

The proof of this theorem is provided in Appendix B.

The GLR stopping rule stops the algorithm at the first time t such that $Z_{s_0}(t) \geq \beta(t, \delta)$, where $(\beta(t, \delta))_{t \in \mathbb{N}}$ is some sequence of thresholds. The stopping time τ_δ induced by this stopping rule can be represented as

$$\tau_\delta = \inf \{t \in \mathbb{N} : Z_{s_0}(t) \geq \beta(t, \delta)\}. \quad (7)$$

As a corollary of Proposition 15 in Kaufmann and Koolen [2021], δ -correctness of any algorithm using this stopping rule is guaranteed if $\beta(t, \delta)$ is set as

$$\beta(t, \delta) = 3 \sum_{\ell \in \mathcal{L}(\mathcal{T})} \log(1 + \log N_\ell(t)) + |\mathcal{L}(\mathcal{T})| \mathcal{C}_{\text{exp}} \left(\frac{\log(1/\delta)}{|\mathcal{L}(\mathcal{T})|} \right) \quad (8)$$

where

$$\mathcal{C}_{\text{exp}}(x) = 2\tilde{h} \left(\frac{h^{-1}(1+x) + \ln(\pi^2/3)}{2} \right), \quad \tilde{h} = \begin{cases} e^{1/h^{-1}(x)} h^{-1}(x) & \text{if } x \geq h(1/\ln(3/2)) \\ \frac{3}{2} (x - \ln \ln \frac{3}{2}) & \text{otherwise} \end{cases}, \quad (9)$$

and $h^{-1}(x)$ is the inverse function of $h(x) = x - \ln x$.

Corollary 4.2 (Corollary of Proposition 15 in Kaufmann and Koolen [2021]). *Any algorithm for Problem 2.2 using the GLR stopping rule with the sequence of thresholds $(\beta(t, \delta))_{t \in \mathbb{N}}$ defined by Eq. (8) ensures*

$$\mathbb{P}(\tau_\delta < \infty, a_{s_0}(\hat{\mu}(\tau_\delta)) \neq a_{s_0}(\mu)) \leq \delta.$$

4.2 Sampling Rules

For the Best Arm Identification problem, Garivier and Kaufmann [2016] proposed two Track-and-Stop sampling rules called C-Tracking and D-Tracking. Our algorithm is based on the D-Tracking sampling rule (Eq. (10)).

$$I(t+1) \leftarrow \begin{cases} \arg \min_{\ell \in \mathcal{L}(\mathcal{T})} N_\ell(t) & \text{if } N_\ell(t) < \sqrt{t} - |\mathcal{L}(\mathcal{T})|/2 \text{ for some } \ell \in \mathcal{L}(\mathcal{T}) \\ \arg \max_{\ell \in \mathcal{L}(\mathcal{T})} tw_\ell^{s_0}(\hat{\mu}(t)) - N_\ell(t) & \text{otherwise} \end{cases} \quad (10)$$

Algorithm 1 Ratio-based D-Tracking for Thresholding MCTS (RD-Tracking-TMCTS)

Require: $\mathcal{T}$: tree, θ : threshold, δ : confidence parameter

Ensure: Return $a_{s_0}(\boldsymbol{\mu}) \in \{\text{'win'}, \text{'lose'}\}$ correctly with prob. at least $1 - \delta$

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1: Draw each leaf node  $\ell \in \mathcal{L}(\mathcal{T})$  once
2: for  $t = 1, 2, \dots$  do
3:    $I(t) \leftarrow \text{Eq. (11)}$ 
4:   Observe reward  $r(t)$  from leaf  $I(t)$  and update  $N_{I(t)}(t)$  and  $\hat{\mu}_{I(t)}(t)$ 
5:   if  $Z_{s_0}(t) \geq \beta(t, \delta)$  then
6:     return  $a_{s_0}(\hat{\boldsymbol{\mu}}(t))$ 

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164 This algorithm tracks the true optimal arm draw proportion through its plug-in estimator and draws
 165 the most under-sampled arm. To ensure consistent tracking, it forces uniform exploration via the first
 166 case of Eq. (10), which makes the empirical means converge to their true values.

167 While it achieves an asymptotically optimal performance, its empirical performance is compromised
 168 by the instability of the selection rule, as we demonstrate later in Section 4.3. In our experiments, the
 169 empirical sample complexity of D-tracking when we adopt Eq. (10) is an order of magnitude worse
 170 than the theoretical bound. To improve the performance, we propose a new sampling rule based on
 171 D-Tracking, **RD-Tracking-TMCTS** (Eq. (11)).

$$I(t+1) \leftarrow \begin{cases} \arg \min_{\ell \in \mathcal{L}(\mathcal{T})} N_\ell(t) & \text{if } N_\ell(t) < \sqrt{t} - |\mathcal{L}(\mathcal{T})|/2 \text{ for some } \ell \in \mathcal{L}(\mathcal{T}) \\ \arg \max_{\ell \in \mathcal{L}(\mathcal{T})} w_\ell^{s_0}(\hat{\boldsymbol{\mu}}(t))/N_\ell(t) & \text{otherwise} \end{cases} \quad (11)$$

172 This sampling rule has the same forced exploration rule as D-Tracking. Instead of selecting an arm
 173 by maximizing $tw_\ell^{s_0}(\hat{\boldsymbol{\mu}}(t)) - N_\ell(t)$, this tracks the target proportion through a ratio-based criterion
 174 involving $w_\ell^{s_0}(\hat{\boldsymbol{\mu}}(t))$ and $N_\ell(t)$. This simple update leads to a several-fold reduction in empirical
 175 sample complexity.

176 4.3 Advantages of the Ratio-based Sampling

177 By Proposition C.4, $w_\ell^{s_0}(\hat{\boldsymbol{\mu}}(t)) = d_{s_0}(\hat{\boldsymbol{\mu}}(t))/d(\hat{\mu}_\ell(t), \theta)$ holds for ℓ with $w_\ell^{s_0}(\hat{\boldsymbol{\mu}}(t)) > 0$. Thus, in
 178 the case with $N_\ell(t) \geq \sqrt{t} - |\mathcal{L}(\mathcal{T})|/2$ for all $\ell \in \mathcal{L}(\mathcal{T})$,

$$I(t+1) \leftarrow \arg \max_{\ell \in \mathcal{L}(\mathcal{T}): w_\ell^{s_0}(\hat{\boldsymbol{\mu}}(t)) > 0} \frac{td_{s_0}(\hat{\boldsymbol{\mu}}(t)) - N_\ell(t)d(\hat{\mu}_\ell(t), \theta)}{d(\hat{\mu}_\ell(t), \theta)} \quad (\text{D-Tracking})$$

$$I(t+1) \leftarrow \arg \max_{\ell \in \mathcal{L}(\mathcal{T}): w_\ell^{s_0}(\hat{\boldsymbol{\mu}}(t)) > 0} \frac{d_{s_0}(\hat{\boldsymbol{\mu}}(t))}{N_\ell(t)d(\hat{\mu}_\ell(t), \theta)} \quad (\text{RD-Tracking})$$

179 hold. This means that RD-Tracking selects ℓ with minimum $N_\ell(t)d(\hat{\mu}_\ell(t), \theta)$. In addition to ℓ with
 180 small $N_\ell(t)d(\hat{\mu}_\ell(t), \theta)$, D-Tracking also prefers ℓ with small $d(\hat{\mu}_\ell(t), \theta)$.

181 For some $L^*(\hat{\boldsymbol{\mu}}(t)) \subseteq \mathcal{L}(\mathcal{T})$, GLR statistic $Z_{s_0}(t)$ (Eq. (6)) is represented as

$$Z_{s_0}(t) = \sum_{\ell \in L^*(\hat{\boldsymbol{\mu}}(t))} N_\ell(t)d(\hat{\mu}_\ell(t), \theta).$$

182 Therefore, if $I(t) \in L^*(\hat{\boldsymbol{\mu}}(t))$ for both D-Tracking and RD-Tracking, Z_{s_0} increases more by increas-
 183 ing $N_\ell(t)$ when $I(t) = \ell$ with larger $d(\hat{\mu}_\ell(t), \theta)$ is selected. This is an intuitive explanation for the
 184 smaller empirical sampling complexity of RD-Tracking.

185 Furthermore, the advantage of the proposed method is also confirmed in experiment (Section 6).

186 4.4 Algorithm

187 Algorithm 1 is the pseudocode of RD-Tracking-TMCTS.

5 Analysis

In Section 5.1, we describe the assumptions necessary for our analysis. In Sections 5.2 and 5.3, we derive an upper bound on the stopping time for Algorithm 1. Furthermore, Section 5.4 shows the computational cost at each time step t for Algorithm 1.

5.1 Assumptions on the True Parameters

We assume that the true parameters μ satisfy the following conditions.

Assumption 5.1. $V_{s_0}(\mu) \neq \theta$.

Assumption 5.2. The optimal proportion $w^{s_0}(\mu)$ is unique.

Assumption 5.1 is required to ensure that the algorithm terminates. Assumption 5.2 is used as a sufficient regularity condition to ensure the asymptotically optimal sample complexity of RD-Tracking-TMCTS. In the case with non-unique optimal proportions, RD-Tracking-TMCTS may be suboptimal, similar to the limitation of D-Tracking (Appendix E of Degenne and Koolen [2019]). We believe that this assumption is not too heavy because non-uniqueness of the optimal proportions is treated as a measure-zero event in many smooth parametric settings. Furthermore, our method remains well-behaved in some cases even when the optimal proportion is not unique (Appendix F). Note that if $c^+(s)$ ($c^-(s)$) is unique for every ‘MAX’ (‘MIN’)-node $s \in S$, Assumption 5.2 holds.

5.2 Almost-Sure Upper Bound of Sample Complexity

The following theorem is the almost sure upper bound of the stopping time for RD-Tracking-TMCTS.

Theorem 5.3. *Using the threshold sequence $(\beta(t, \delta))_{t \in \mathbb{N}}$ defined by Eq. (8), RD-Tracking-TMCTS guarantees that for all $\delta \in (0, 1)$, $\mathbb{P}(\tau_\delta < +\infty) = 1$ and*

$$\mathbb{P}_\mu \left(\limsup_{\delta \rightarrow 0} \frac{\tau_\delta}{\log(1/\delta)} \leq \frac{1}{d_{s_0}(\mu)} \right) = 1.$$

The proof of this theorem is provided in Appendix C.

5.3 Upper Bound of Expected Sample Complexity

The following theorem states that RD-Tracking-TMCTS achieves asymptotically optimal expected sample complexity.

Theorem 5.4. *For RD-Tracking-TMCTS with the threshold sequence $(\beta(t, \delta))_{t \in \mathbb{N}}$ defined by Eq. (8), expected stopping time $\mathbb{E}_\mu[\tau_\delta]$ is asymptotically upper-bounded by*

$$\limsup_{\delta \rightarrow +0} \frac{\mathbb{E}_\mu[\tau_\delta]}{\log(1/\delta)} \leq \frac{1}{d_{s_0}(\mu)}.$$

The proof of this theorem is provided in Appendix D.

Note that, given the matching lower bound, Theorem 5.4, which bounds the expected stopping time, is slightly stronger than Theorem C.1, which bounds the stopping time with probability 1. This is because a sequence of random variables that converges to a constant with probability 1 can have a larger (or even diverging) expected value than the constant.

5.4 Time Complexity

Consider the time complexity of calculating $I(t)$ in Eq. (11). Forced exploration can be implemented in $O(\log |\mathcal{L}(\mathcal{T})|)$ time using a heap for $\{N_\ell(t) \mid \ell \in \mathcal{L}(\mathcal{T})\}$. The ratio-based sampling must update the weight vector $w^{s_0}(\hat{\mu}(t-1))$ after each draw and choose its maximizer. The straightforward implementation of the sampling requires recomputation of the entire weight vector $w^{s_0}(\mu)$ at each step, which takes $O(|S|)$ time using Recursive Formula 1. However, if $\mathcal{T}$ is balanced, the computation time $O(\log |S|)$ is achievable by recursively calculating $\text{RD}_s(t) = (1/(d_{I_s(t)}(\hat{\mu}(t))N_{I_s(t)}(t)), I_s(t))$, where $I_s(t) = \arg \max_{\ell \in \mathcal{D}(s)} w_\ell^s(\hat{\mu}(t))/N_\ell(t)$. Note that

227 $\arg \max_{\ell \in \mathcal{D}(s)} w_\ell^s(\hat{\mu}(t))/N_\ell(t) = \arg \max_{\ell \in \mathcal{D}(s)} 1/(d_s(\hat{\mu}(t))N_\ell(t))$ by Proposition C.4. (See Recursive For-
 228 mula 2 and Theorem E.1 in Appendix E.)

229 Let $\mathcal{D}(\mathcal{T})$ be the depth of $\mathcal{T}$, and let $\mathcal{W}(\mathcal{T})$ be the maximum number of children of $s \in S$, that is,
 230 $\max_{s \in S} |\mathcal{C}(s)|$. Then, the following theorem holds.

231 **Theorem 5.5.** *RD-Tracking-TMCTS (Algorithm 1) runs in $O(|S|)$ space and $O(\mathcal{D}(\mathcal{T}) \log \mathcal{W}(\mathcal{T}))$
 232 amortized time per round.*

233 Since $|\mathcal{L}(\mathcal{T})| \leq \mathcal{W}(\mathcal{T})^{\mathcal{D}(\mathcal{T})}$, we have $\log |\mathcal{L}(\mathcal{T})| \leq \mathcal{D}(\mathcal{T}) \log \mathcal{W}(\mathcal{T})$. If the tree is balanced, that
 234 is, if $|\mathcal{L}(\mathcal{T})| = \Theta(\mathcal{W}(\mathcal{T})^{\mathcal{D}(\mathcal{T})})$ holds, then $\mathcal{D}(\mathcal{T}) \log \mathcal{W}(\mathcal{T}) = \Theta(\log |\mathcal{L}(\mathcal{T})|)$ (and also $\Theta(\log |S|)$)
 235 holds.

236 Theorem 5.5 is achieved through the careful construction of data structures and algorithmic imple-
 237 mentations, which we elaborate on in Appendix E.

238 6 Experimental Results

239 We conduct experiments on instances where the label of the root node s_0 satisfies $L(s_0) = \text{'MAX'}$.

240 To validate the effectiveness of our method, RD-Tracking-TMCTS, we compare it with several base-
 241 line algorithms. Specifically, we use D-Tracking [Garivier and Kaufmann, 2016], C-Tracking [Gariv-
 242 ier and Kaufmann, 2016], UGapE-MCTS [Kaufmann and Koolen, 2017], LUCB-micro [Huang et al.,
 243 2017], and round-robin (= uniform sampling) as baselines.

244 D-Tracking and C-Tracking are instances of the Track-and-Stop algorithm. To apply these algorithms
 245 to our Thresholding MCTS problem, we made them select a leaf following the plug-in estimator of
 246 the optimal weight $w(\hat{\mu}(t))$, which leads to convergence of the sampling proportion to the optimal
 247 weight $w(\mu)$. For fairness, we employ the same stopping rule as in the proposed method and use the
 248 same confidence function $\beta(t, \delta) = \log(1 + \log t) + \log(1/\delta)$. Since these methods ensure that the
 249 selection proportions converge to $w(\mu(t))$, they satisfy the conditions of Theorem C.1.

250 UGapE-MCTS and LUCB-micro are confidence-interval-based algorithms designed for the Best Arm
 251 Identification problem. Since these algorithms identify $s^* = \arg \max_{c \in \mathcal{C}(s_0)} V_c(\mu)$, we modify them to

252 instead return whether the estimated value of s^* is at least θ . For UGapE-MCTS, we set the tolerance
 253 parameter ϵ to zero, thereby requiring the algorithm to identify the exact s^* . For fairness, we define
 254 the confidence interval at the leaf node ℓ as $\left[\hat{\mu}_\ell(t) - \sqrt{\frac{\beta(N_\ell(t), \delta)}{2N_\ell(t)}}, \hat{\mu}_\ell(t) + \sqrt{\frac{\beta(N_\ell(t), \delta)}{2N_\ell(t)}} \right]$, and we use
 255 the same function $\beta(t, \delta)$ as that employed in the proposed method.

256 The round-robin algorithm pulls each leaf in a fixed cyclic order. We employ the same stopping
 257 rule as in the proposed method and use the same confidence function $\beta(t, \delta)$. Once the condition
 258 $Z_{s_0}(t) \geq \beta(t, \delta)$ is met, the algorithm terminates and outputs $a_{s_0}(\hat{\mu})$.

259 We conducted experiments on randomly generated 500 trees, where each node has a branching factor
 260 randomly chosen from $\{2, 3, 4, 5\}$, and each node has children with probability 0.5 up to a maximum
 261 depth of 5. We set nodes at even depths as MAX nodes and those at odd depths as MIN nodes. Each
 262 μ_ℓ is randomly sampled from a uniform distribution over $[0, 1]$. To ensure a fair comparison with
 263 Best Arm Identification methods, we constrained the value of the second-best depth-1 node to θ . This
 264 setting makes the gap between the best and second-best nodes equal to the gap between the best node
 265 and the threshold.

266 We performed experiments with $\delta = 10^{-1}, 10^{-2}, \dots, 10^{-6}, 10^{-8}, \dots, 10^{-20}$. For each method and
 267 each value of δ , we ran 10,000 times (20 iteration $\times$ 500 trees) and measured the average, 25%, and
 268 75% percentile of the stopping time. Figure 1 shows the ratio of the stopping time to its average
 269 asymptotic lower bound $\mathbb{E}_{\mu \in \text{test cases}} [\log(1/\delta)/d_{s_0}(\mu)]$ averaged over 10,000 iterations. The shaded
 270 region enclosed by the dotted lines represent a 25% and 75% percentile of the stopping time.

271 For all δ , RD-Tracking-TMCTS attained the lowest average stopping time among the compared
 272 methods. Since the experiments cover a wide range of confidence levels, from $\delta = 0.1$ to $\delta = 10^{-20}$,
 273 the results demonstrate that the our method remains effective both in practical finite- δ settings and in
 274 extremely small- δ regimes that are often relevant in asymptotic analysis.

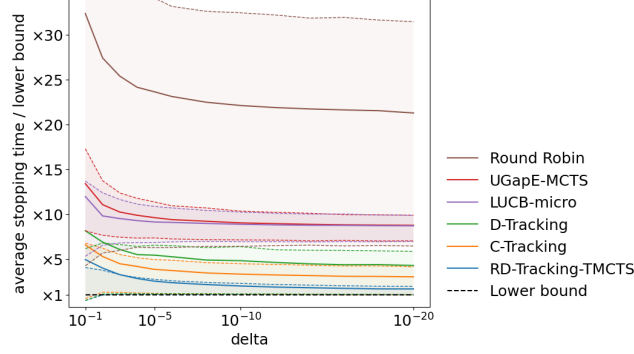


Figure 1: Ratio of the average stopping time to the lower bound for different values of δ . The shaded region enclosed by the dotted lines represent a 25% and 75% percentile.

Although RD-Tracking-TMCTS, as well as C-Tracking and D-Tracking, are all theoretically guaranteed to be asymptotically optimal, the rates at which their stopping times approach the lower bound differ substantially. The ratios of the average stopping time to the lower bound for UGapE-MCTS, LUCB-micro, and round-robin do not seem to approach 1 as $\delta \rightarrow 0$, indicating that these methods are not asymptotically optimal. Moreover, we did not observe any incorrect outputs in all trials for any setting, which is consistent with δ -correctness.

We also evaluate the robustness of the proposed method for varying θ , including cases where θ is close to $V_{s_0}(\mu)$. While the average stopping time itself changes with the intrinsic difficulty of each instance, the ratio to the theoretical lower bound remains stable across different values of θ . The detailed results are provided in Appendix G.

7 Good Action Identification Problem

Up to now, we have focused on the thresholding bandit problem, where the goal is to determine whether the value of the root node is at least a given threshold θ . When $L(s_0) = \text{'MAX'}$ ('MIN'), identifying $a_{s_0}(\mu)$ is equivalent to determining whether there exists some $c \in \mathcal{C}(s_0)$ such that $V_c(\mu) \geq \theta$ ($V_c(\mu) < \theta$). A natural question is to ask whether our framework can be applied to find a good action at the root node s_0 with $L(s_0) = \text{'MAX'}$ ('MIN'), i.e., the child node whose value is at least (less than) θ . We call this problem the good action identification (GAI) problem and answer this question affirmatively.

Despite the fact that the GAI problem is intrinsically more challenging than the thresholding problem, its asymptotic upper bound is identical to that of the thresholding problem. Furthermore, a suitable modification of RD-Tracking-TMCTS yields an algorithm whose expected stopping time satisfies

$$\limsup_{\delta \rightarrow +0} \frac{\mathbb{E}[\tau_\delta]}{\log(1/\delta)} \leq \frac{1}{d_{s_0}(\mu)}$$

and achieves asymptotic optimality. We defer the details to Appendix H.

8 Limitations and Future Work

- Our analysis limited for fixed-tree setting and does not covers dynamic tree expansion.
- Our optimality guarantee is asymptotic and does not yield a non-asymptotic stopping-time bound; nevertheless, the experiments suggest that the RD-Tracking-TMCTS performs well in the finite-confidence regimes. (Section 6)
- As future work, it would be interesting to investigate whether the Best Action Identification problem can be solved by repeatedly applying the GAI formulation with varying thresholds θ . Because the threshold-dependent analysis depends heavily on the distance between the threshold and the best-valued action, we expect this approach to be especially effective in the ϵ -best arm identification setting with a very large number of arms.

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449 A Proof of Theorem 3.2 (Calculation of Optimal Weights)

450 *Proof.* We prove by mathematical induction on the height of the tree node s .

451 Consider a height-0 tree node s . In this case, node s is a leaf node itself, thus $w_s(\mu) = 1$. In this
452 case,

$$d_s(\mu) = \max_{\mathbf{w} \in \Delta_{\mathcal{D}(s)}} \inf_{\lambda \in \text{Alt}_s(\mu)} \sum_{\ell \in \mathcal{D}(s)} w_\ell d(\mu_\ell, \lambda_\ell) = d(\mu_s, \theta)$$

453 which coincides with weights $w_s^s(\mu)$ and $|\tilde{d}_s(\mu)|$ calculated by Recursive Formula 1. Furthermore,

$$\tilde{d}_s(\mu) = \begin{cases} d(\mu_s, \theta) \geq 0 & (a_s(\mu) = \text{'win'}) \\ -d(\mu_s, \theta) < 0 & (a_s(\mu) = \text{'lose'}). \end{cases}$$

454 Assume that $w^s(\mu)$ is calculated by Recursive Formula 1 and

$$d_s(\mu) = \max_{w \in \Delta_{\mathcal{D}(s)}} \inf_{\lambda \in \text{Alt}_s(\mu)} \sum_{\ell \in \mathcal{D}(s)} w_\ell d(\mu_\ell, \lambda_\ell) \text{ and}$$

$$\tilde{d}_s(\mu) = \begin{cases} d_s(\mu) \geq 0 & (a_s(\mu) = \text{'win'}) \\ -d_s(\mu) < 0 & (a_s(\mu) = \text{'lose'}) \end{cases}$$

455 hold for a tree node s with the height at most k . Consider a height- $(k+1)$ tree node s .

456 Assume that node s is a MAX node and $A_s(\mu) \neq \emptyset$, that is, $a_s(\mu) = \text{'win'}$. Then,

$$\inf_{\lambda \in \text{Alt}_s(\mu)} \sum_{\ell \in \mathcal{D}(s)} w_\ell d(\mu_\ell, \lambda_\ell) = \sum_{c \in \mathcal{C}(s), V_c(\mu) \geq \theta} W_c \inf_{\lambda \in \text{Alt}_c(\mu)} \sum_{\ell \in \mathcal{D}(c)} w'_\ell d(\mu_\ell, \lambda_\ell)$$

457 for any $w \in \Delta_{\mathcal{D}(s)}$, where $W_c = \sum_{\ell \in \mathcal{D}(c)} w_\ell$ and $w'_\ell = w_\ell / W_c$. Let optimal $(W_c)_{c \in \mathcal{C}(s)}$ denote

458 $(W_c^*)_{c \in \mathcal{C}(s)}$. Thus,

$$\begin{aligned} \max_{w \in \Delta_{\mathcal{D}(s)}} \inf_{\lambda \in \text{Alt}_s(\mu)} \sum_{\ell \in \mathcal{D}(s)} w_\ell d(\mu_\ell, \lambda_\ell) &= \max_{w \in \Delta_{\mathcal{D}(s)}} \sum_{c \in \mathcal{C}(s), V_c(\mu) \geq \theta} W_c \max_{w' \in \Delta_{\mathcal{D}(c)}} \inf_{\lambda \in \text{Alt}_c(\mu)} \sum_{\ell \in \mathcal{D}(c)} w'_\ell d(\mu_\ell, \lambda_\ell) \\ &= \max_{w \in \Delta_{\mathcal{D}(s)}} \sum_{c \in \mathcal{C}(s), V_c(\mu) \geq \theta} W_c \tilde{d}_c(\mu) \\ &= \sum_{c \in \mathcal{C}(s), V_c(\mu) \geq \theta} W_c^* \tilde{d}_c(\mu) \end{aligned}$$

459 where

$$W_c^* = \begin{cases} 1 & \left(c = \arg \max_{c' \in A_s(\mu)} \tilde{d}_{c'}(\mu) \right) \\ 0 & (\text{otherwise}). \end{cases}$$

460 Therefore,

$$w_\ell^s(\mu) = \begin{cases} w_\ell^c(\mu) & (c = \arg \max_{c' \in A_s(\mu)} \tilde{d}_{c'}(\mu), \ell \in \mathcal{D}(c)) \\ 0 & (\text{otherwise}), \end{cases}$$

461 which coincides with weights $w_\ell^s(\mu)$ ($\ell \in \mathcal{D}(s)$) calculated by Recursive Formula 1. In this case,

$$\tilde{d}_s(\mu) = \max_{c \in \mathcal{C}(s)} \tilde{d}_c(\mu) = \sum_{c \in \mathcal{C}(s), V_c(\mu) \geq \theta} W_c^* \tilde{d}_c(\mu) = \max_{w \in \Delta_{\mathcal{D}(s)}} \inf_{\lambda \in \text{Alt}_s(\mu)} \sum_{\ell \in \mathcal{D}(s)} w_\ell d(\mu_\ell, \lambda_\ell)$$

462 holds.

463 Assume that node s is a MAX node and $A_s(\mu) = \emptyset$, that is, $a_s(\mu) = \text{'lose'}$. Then,

$$\inf_{\lambda \in \text{Alt}_s(\mu)} \sum_{\ell \in \mathcal{D}(s)} w_\ell d(\mu_\ell, \lambda_\ell) = \min_{\ell \in \mathcal{C}(s)} W_c \inf_{\lambda \in \text{Alt}_c(\mu)} \sum_{\ell \in \mathcal{D}(c)} w'_\ell d(\mu_\ell, \lambda_\ell)$$

464 for any $w \in \Delta_{\mathcal{D}(s)}$, where $W_c = \sum_{\ell \in \mathcal{D}(c)} w_\ell$ and $w'_\ell = w_\ell / W_c$. Thus,

$$\begin{aligned} \max_{w \in \Delta_{\mathcal{D}(s)}} \inf_{\lambda \in \text{Alt}_s(\mu)} \sum_{\ell \in \mathcal{D}(s)} w_\ell d(\mu_\ell, \lambda_\ell) &= \max_{w \in \Delta_{\mathcal{D}(s)}} \min_{c \in \mathcal{C}(s)} W_c \max_{w' \in \Delta_{\mathcal{D}(c)}} \inf_{\lambda \in \text{Alt}_c(\mu)} \sum_{\ell \in \mathcal{D}(c)} w'_\ell d(\mu_\ell, \lambda_\ell) \\ &= \max_{w \in \Delta_{\mathcal{D}(s)}} \min_{c \in \mathcal{C}(s)} W_c (-\tilde{d}_c(\mu)) \end{aligned}$$

465 holds. Let optimal $(W_c)_{c \in \mathcal{C}(s)}$ denote $(W_c^*)_{c \in \mathcal{C}(s)}$. Then,

$$W_c^* = \frac{1/(-\tilde{d}_c(\mu))}{\sum_{c' \in \mathcal{C}(s)} 1/(-\tilde{d}_{c'}(\mu))} = \frac{1/\tilde{d}_c(\mu)}{\sum_{c' \in \mathcal{C}(s)} 1/\tilde{d}_{c'}(\mu)}$$

466 holds, which means

$$w_\ell^s(\boldsymbol{\mu}) = \frac{1/\tilde{d}_c(\boldsymbol{\mu})}{\sum_{c' \in \mathcal{C}(s)} 1/\tilde{d}_{c'}(\boldsymbol{\mu})} w_\ell^c(\boldsymbol{\mu})$$

467 for $c \in \mathcal{C}(s)$ and $\ell \in \mathcal{D}(c)$. These weights $w_\ell^s(\boldsymbol{\mu})$ coincide with the weights $w_\ell^s(\boldsymbol{\mu})$ ($\ell \in \mathcal{D}(s)$)
 468 calculated by Recursive Formula 1. In this case,

$$-\tilde{d}_s(\boldsymbol{\mu}) = -\frac{1}{\sum_{c' \in \mathcal{C}(s)} 1/\tilde{d}_{c'}(\boldsymbol{\mu})} = W_c^*(-\tilde{d}_c(\boldsymbol{\mu})) = \max_{\mathbf{w} \in \Delta_{\mathcal{D}(s)}} \inf_{\boldsymbol{\lambda} \in \text{Alt}_s(\boldsymbol{\mu})} \sum_{\ell \in \mathcal{D}(s)} w_\ell d(\mu_\ell, \lambda_\ell)$$

469 holds for any $c \in \mathcal{C}(s)$.

470 Assume that node s is a MIN node and $A_s(\boldsymbol{\mu}) = \mathcal{C}(s)$, that is, $a_s(\boldsymbol{\mu}) = \text{'win'}$. Then,

$$\inf_{\boldsymbol{\lambda} \in \text{Alt}_s(\boldsymbol{\mu})} \sum_{\ell \in \mathcal{D}(s)} w_\ell d(\mu_\ell, \lambda_\ell) = \min_{\ell \in \mathcal{C}(s)} W_c \inf_{\boldsymbol{\lambda} \in \text{Alt}_c(\boldsymbol{\mu})} \sum_{\ell \in \mathcal{D}(c)} w'_\ell d(\mu_\ell, \lambda_\ell)$$

471 for any $\mathbf{w} \in \Delta_{\mathcal{D}(s)}$, where $W_c = \sum_{\ell \in \mathcal{D}(c)} w_\ell$ and $w'_\ell = w_\ell/W_c$. Thus,

$$\begin{aligned} \max_{\mathbf{w} \in \Delta_{\mathcal{D}(s)}} \inf_{\boldsymbol{\lambda} \in \text{Alt}_s(\boldsymbol{\mu})} \sum_{\ell \in \mathcal{D}(s)} w_\ell d(\mu_\ell, \lambda_\ell) &= \max_{\mathbf{w} \in \Delta_{\mathcal{C}(s)}} \min_{c \in \mathcal{C}(s)} W_c \max_{\mathbf{w}' \in \Delta_{\mathcal{D}(c)}} \inf_{\boldsymbol{\lambda} \in \text{Alt}_c(\boldsymbol{\mu})} \sum_{\ell \in \mathcal{D}(c)} w'_\ell d(\mu_\ell, \lambda_\ell) \\ &= \max_{\mathbf{w} \in \Delta_{\mathcal{C}(s)}} \min_{c \in \mathcal{C}(s)} W_c \tilde{d}_c(\boldsymbol{\mu}) \end{aligned}$$

472 holds. Let optimal $(W_c)_{c \in \mathcal{C}(s)}$ denote $(W_c^*)_{c \in \mathcal{C}(s)}$. Then,

$$W_c^* = \frac{1/\tilde{d}_c(\boldsymbol{\mu})}{\sum_{c' \in \mathcal{C}(s)} 1/\tilde{d}_{c'}(\boldsymbol{\mu})}$$

473 holds, which means

$$w_\ell^s(\boldsymbol{\mu}) = \frac{1/\tilde{d}_c(\boldsymbol{\mu})}{\sum_{c' \in \mathcal{C}(s)} 1/\tilde{d}_{c'}(\boldsymbol{\mu})} w_\ell^c(\boldsymbol{\mu})$$

474 for $c \in \mathcal{C}(s)$ and $\ell \in \mathcal{D}(c)$. These weights $w_\ell^s(\boldsymbol{\mu})$ coincide with the weights $w_\ell^s(\boldsymbol{\mu})$ ($\ell \in \mathcal{D}(s)$)
 475 calculated by Recursive Formula 1. In this case,

$$\tilde{d}_s(\boldsymbol{\mu}) = \frac{1}{\sum_{c' \in \mathcal{C}(s)} 1/\tilde{d}_{c'}(\boldsymbol{\mu})} = W_c^* \tilde{d}_c(\boldsymbol{\mu}) = \max_{\mathbf{w} \in \Delta_{\mathcal{D}(s)}} \inf_{\boldsymbol{\lambda} \in \text{Alt}_s(\boldsymbol{\mu})} \sum_{\ell \in \mathcal{D}(s)} w_\ell d(\mu_\ell, \lambda_\ell)$$

476 holds for any $c \in \mathcal{C}(s)$.

477 Assume that node s is a MIN node and $A_s(\boldsymbol{\mu}) \neq \mathcal{C}(s)$, that is, $a_s(\boldsymbol{\mu}) = \text{'lose'}$. Then,

$$\inf_{\boldsymbol{\lambda} \in \text{Alt}_s(\boldsymbol{\mu})} \sum_{\ell \in \mathcal{D}(s)} w_\ell d(\mu_\ell, \lambda_\ell) = \sum_{c \in \mathcal{C}(s), V_c(\boldsymbol{\mu}) < \theta} W_c \inf_{\boldsymbol{\lambda} \in \text{Alt}_c(\boldsymbol{\mu})} \sum_{\ell \in \mathcal{D}(c)} w'_\ell d(\mu_\ell, \lambda_\ell)$$

478 for any $\mathbf{w} \in \Delta_{\mathcal{D}(s)}$, where $W_c = \sum_{\ell \in \mathcal{D}(c)} w_\ell$ and $w'_\ell = w_\ell/W_c$. Let optimal $(W_c)_{c \in \mathcal{C}(s)}$ denote

479 $(W_c^*)_{c \in \mathcal{C}(s)}$. Thus,

$$\begin{aligned} \max_{\mathbf{w} \in \Delta_{\mathcal{D}(s)}} \inf_{\boldsymbol{\lambda} \in \text{Alt}_s(\boldsymbol{\mu})} \sum_{\ell \in \mathcal{D}(s)} w_\ell d(\mu_\ell, \lambda_\ell) &= \max_{\mathbf{w} \in \Delta_{\mathcal{C}(s)}} \sum_{c \in \mathcal{C}(s), V_c(\boldsymbol{\mu}) < \theta} W_c \max_{\mathbf{w}' \in \Delta_{\mathcal{D}(c)}} \inf_{\boldsymbol{\lambda} \in \text{Alt}_c(\boldsymbol{\mu})} \sum_{\ell \in \mathcal{D}(c)} w'_\ell d(\mu_\ell, \lambda_\ell) \\ &= \max_{\mathbf{w} \in \Delta_{\mathcal{C}(s)}} \sum_{c \in \mathcal{C}(s), V_c(\boldsymbol{\mu}) < \theta} W_c(-\tilde{d}_c(\boldsymbol{\mu})) \\ &= \sum_{c \in \mathcal{C}(s), V_c(\boldsymbol{\mu}) < \theta} W_c^*(-\tilde{d}_c(\boldsymbol{\mu})) \end{aligned}$$

480 where

$$W_c^* = \begin{cases} 1 & \left(c = \arg \max_{c' \in \mathcal{C}(s) \setminus A_s(\boldsymbol{\mu})} (-\tilde{d}_{c'}(\boldsymbol{\mu})) \right) \\ 0 & (\text{otherwise}). \end{cases}$$

Therefore,

$$w_\ell^s(\boldsymbol{\mu}) = \begin{cases} w_\ell^c(\boldsymbol{\mu}) & (c = \arg \min_{c' \in \mathcal{C}(s) \setminus A_s(\boldsymbol{\mu})} \tilde{d}_{c'}(\boldsymbol{\mu}), \ell \in \mathcal{D}(c)) \\ 0 & (\text{otherwise}), \end{cases}$$

which coincides with weights $w_\ell^s(\boldsymbol{\mu})$ ($\ell \in \mathcal{D}(s)$) calculated by Recursive Formula 1. In this case,

$$-\tilde{d}_s(\boldsymbol{\mu}) = -\min_{c \in \mathcal{C}(s)} \tilde{d}_c(\boldsymbol{\mu}) = \max_{c \in \mathcal{C}(s)} (-\tilde{d}_c(\boldsymbol{\mu})) = \sum_{c \in \mathcal{C}(s), V_c < \theta} W_c^*(-\tilde{d}_c(\boldsymbol{\mu})) = \max_{\mathbf{w} \in \Delta_{\mathcal{D}(s)}} \inf_{\boldsymbol{\lambda} \in \text{Alt}_s(\boldsymbol{\mu})} \sum_{\ell \in \mathcal{D}(s)} w_\ell d(\mu_\ell, \lambda_\ell)$$

holds.

484

□

B Proof of Theorem 4.1 (Calculation of GLR Statistics)

Proof. We prove the theorem by mathematical induction on the height of the tree node s . Consider a height-0 tree node s . Then, GLR $Z_s(t)$ can be calculated as

$$\begin{aligned} Z_s(t) &= \inf_{\boldsymbol{\lambda} \in \text{Alt}_s(\hat{\boldsymbol{\mu}}(t))} \sum_{\ell \in \mathcal{D}(s)} N_\ell(t) d(\hat{\mu}_\ell(t), \lambda_\ell) \\ &= N_s(t) d(\hat{\mu}_s(t), \theta) \\ &= \begin{cases} \tilde{Z}_s(t) & (\hat{\mu}_s \geq \theta) \\ -\tilde{Z}_s(t) & (\hat{\mu}_s < \theta) \end{cases} \end{aligned}$$

which coincides with the absolute value of $\tilde{Z}_s(t)$ calculated by Recursive Formula 1.

Assume that the GLR $Z_s(t)$ defined by Eq. (6) can be calculated through Recursive Formula 1 for any height- k tree node s . Consider a height- $(k+1)$ tree node s .

In the case with $a_s(\hat{\boldsymbol{\mu}}(t)) = \text{'win'}$,

$$\begin{aligned} Z_s(t) &= \inf_{\boldsymbol{\lambda} \in \text{Alt}_s(\hat{\boldsymbol{\mu}}(t))} \sum_{\ell \in \mathcal{D}(s)} N_\ell(t) d(\hat{\mu}_\ell(t), \lambda_\ell) \\ &= \begin{cases} \sum_{c \in A_s(\hat{\boldsymbol{\mu}})} \inf_{\boldsymbol{\lambda} \in \text{Alt}_c(\hat{\boldsymbol{\mu}}(t))} \sum_{\ell \in \mathcal{D}(c)} N_\ell(t) d(\mu_\ell, \lambda_\ell) = \sum_{c \in A_s(\hat{\boldsymbol{\mu}}(t))} \tilde{Z}_c(t) = \tilde{Z}_s(t) & (L(s) = \text{'MAX'}) \\ \min_{c \in \mathcal{C}(s)} \inf_{\boldsymbol{\lambda} \in \text{Alt}_c(\hat{\boldsymbol{\mu}}(t))} \sum_{\ell \in \mathcal{D}(c)} N_\ell(t) d(\mu_\ell, \lambda_\ell) = \min_{c \in \mathcal{C}(s)} \tilde{Z}_c(t) = \tilde{Z}_s(t) & (L(s) = \text{'MIN'}) \end{cases} \end{aligned}$$

holds, where $\tilde{Z}_c(t) = \inf_{\boldsymbol{\lambda} \in \text{Alt}_c(\hat{\boldsymbol{\mu}}(t))} \sum_{\ell \in \mathcal{D}(c)} N_\ell d(\mu_\ell, \lambda_\ell)$ holds for c with $a_c(\hat{\boldsymbol{\mu}}(t)) = \text{'win'}$ by the induction assumption. Thus, the GLR $Z_s(t)$ is the absolute value of $\tilde{Z}_s(t)$ calculated by Recursive Formula 1.

In the case with $a_s(\hat{\boldsymbol{\mu}}(t)) = \text{'lose'}$,

$$\begin{aligned} Z_s(t) &= \inf_{\boldsymbol{\lambda} \in \text{Alt}_s(\hat{\boldsymbol{\mu}}(t))} \sum_{\ell \in \mathcal{D}(s)} N_\ell(t) d(\hat{\mu}_\ell(t), \lambda_\ell) \\ &= \begin{cases} \min_{c \in \mathcal{C}(s)} \inf_{\boldsymbol{\lambda} \in \text{Alt}_c(\hat{\boldsymbol{\mu}}(t))} \sum_{\ell \in \mathcal{D}(c)} N_\ell d(\mu_\ell, \lambda_\ell) = \min_{c \in \mathcal{C}(s)} (-\tilde{Z}_c(t)) = -\max_{c \in \mathcal{C}(s)} \tilde{Z}_c(t) = -\tilde{Z}_s(t) & (L(s) = \text{'MAX'}) \\ \sum_{c \in \mathcal{C}(s) \setminus A_s(\hat{\boldsymbol{\mu}}(t))} \inf_{\boldsymbol{\lambda} \in \text{Alt}_c(\hat{\boldsymbol{\mu}}(t))} \sum_{\ell \in \mathcal{D}(c)} N_\ell d(\mu_\ell, \lambda_\ell) = \sum_{c \in \mathcal{C}(s) \setminus A_s(\hat{\boldsymbol{\mu}}(t))} (-\tilde{Z}_c(t)) = -\tilde{Z}_s(t) & (L(s) = \text{'MIN'}) \end{cases} \end{aligned}$$

holds, where $-\tilde{Z}_c(t) = \inf_{\boldsymbol{\lambda} \in \text{Alt}_c(\hat{\boldsymbol{\mu}}(t))} \sum_{\ell \in \mathcal{D}(c)} N_\ell d(\mu_\ell, \lambda_\ell)$ holds for c with $a_c(\hat{\boldsymbol{\mu}}(t)) = \text{'lose'}$ by the induction assumption. Thus, the GLR $Z_s(t)$ is the absolute value of $\tilde{Z}_s(t)$ calculated by Recursive Formula 1. □

C Proofs for Almost-Sure Upper Bound of Stopping Time

Theorem 5.3 is derived straightforwardly from the following two lemmas.

The first lemma is the almost sure upper bound of the stopping time for the class of algorithms whose sampling proportion of leaf nodes converges to the optimal proportion.

Lemma C.1. *Using GLR stopping rule with the threshold $\beta(t, \delta)$ defined by Eq. (8), and any sampling rule ensuring that for every leaf $\ell \in \mathcal{L}(T)$, $N_\ell(t)/t$ converges almost surely to $w_\ell^{s_0}(\mu)$, any algorithm for Problem 2.2 guarantees that for all $\delta \in (0, 1)$, $\mathbb{P}(\tau_\delta < +\infty) = 1$ and*

$$\mathbb{P}_\mu \left(\limsup_{\delta \rightarrow 0} \frac{\tau_\delta}{\log(1/\delta)} \leq \frac{1}{d_{s_0}(\mu)} \right) = 1.$$

The proof of this theorem is provided in Sec. C.1.

The next lemma guarantees that the sampling ratio of RD-Tracking-TMCTS converges to the optimal ratio. (See Sec. C.3 for the proof.)

Lemma C.2. *RD-Tracking-TMCTS sampling rule satisfies*

$$\mathbb{P} \left(\lim_{t \rightarrow \infty} \frac{N_\ell(t)}{t} = w_\ell^{s_0}(\mu) \right) = 1.$$

C.1 Proof of Lemma C.1 (Almost-Sure Upper Bound of Stopping Time)

Lemma C.5 and Propositions C.6 and C.7 are used to prove Lemma C.8, which is needed to prove Lemma C.1. First, we prove Propositions C.3 and C.4 to prove Lemma C.5.

Proposition C.3. *For $\ell \in \mathcal{L}(\mathcal{T})$ with $w_\ell^{s_0}(\mu) > 0$,*

$$\mu_\ell \begin{cases} > \theta & (a_{s_0}(\mu) = \text{'win'}) \\ < \theta & (a_{s_0}(\mu) = \text{'lose'}) \end{cases}$$

holds.

Proof. By Theorem 3.2, we prove this proposition for $w^s(\mu)$ calculated using Recursive Formulae 1.

Let $s_0, s_1, \dots, s_k = \ell$ be the node sequence on the path from node s_0 to node ℓ . Consider the case with $a_{s_0}(\mu) = \text{'win'}$. Then, $V_{s_0}(\mu) > \theta$ by Assumption 5.1. In this case, Condition $w_\ell^{s_0}(\mu) > 0$ implies that, for $i = 0, 1, \dots, k-1$, $V_{s_i}(\mu) = V_{s_{i+1}}(\mu)$ for node s_i with $L(s_i) = \text{'MAX'}$ and $V_{s_i}(\mu) \leq V_{s_{i+1}}(\mu)$ for node s_i with $L(s_i) = \text{'MIN'}$. Thus, $\theta < V_{s_0}(\mu) \leq V_{s_1}(\mu) \leq \dots \leq V_{s_k}(\mu) = V_\ell(\mu) = \mu_\ell$ holds. In the case with $a_{s_0}(\mu) = \text{'lose'}$, $V_{s_0}(\mu) < \theta$ holds. Condition $w_\ell^{s_0}(\mu) > 0$ implies that, for $i = 0, 1, \dots, k-1$, $V_{s_i}(\mu) = V_{s_{i+1}}(\mu)$ for node s_i with $L(s_i) = \text{'MIN'}$ and $V_{s_i}(\mu) \geq V_{s_{i+1}}(\mu)$ for node s_i with $L(s_i) = \text{'MAX'}$. Thus, $\theta > V_{s_0}(\mu) \geq V_{s_1}(\mu) \geq \dots \geq V_{s_k}(\mu) = V_\ell(\mu) = \mu_\ell$ holds. $\square$

Proposition C.4. *For $\ell \in \mathcal{L}(\mathcal{T})$ with $w_\ell^{s_0}(\mu) > 0$,*

$$w_\ell^s(\mu) = \frac{d_s(\mu)}{d(\mu_\ell, \theta)} \tag{12}$$

holds for $s = s_0, s_1, \dots, s_k (= \ell)$, where $s_0, s_1, \dots, s_k$ is the node sequence on the path from node s_0 to node ℓ .

Proof. By Theorem 3.2, Eq. (12) can be proved for $w^s(\mu)$ and $d_s(\mu)$ calculated using Recursive Formula 1.

We prove Eq. (12) by mathematical induction.

Consider the case with $s = s_k$. By proposition C.3 and Recursive Formula 1,

$$d_s(\mu) = |\tilde{d}_s(\mu)| = d(\mu_\ell, \theta).$$

Thus, in this case,

$$w_\ell^s(\mu) = 1 = \frac{d_s(\mu)}{d(\mu_\ell, \theta)}$$

holds.

Assume that Eq. (12) holds for $s = s_i$ ($1 \leq i \leq k$). Consider the case with $a_{s_0}(\mu) = \text{'win'}$. If $L(s_{i-1}) = \text{'MAX'}$, $w_\ell^{s_0}(\mu) > 0$ implies $w_\ell^{s_{i-1}}(\mu) = w_\ell^{s_i}(\mu)$ and $d_{s_{i-1}}(\mu) = |\tilde{d}_{s_{i-1}}(\mu)| = |\tilde{d}_{s_i}(\mu)| = d_{s_i}(\mu)$, thus

$$w_\ell^{s_{i-1}}(\mu) = w_\ell^{s_i}(\mu) = \frac{d_{s_i}(\mu)}{d(\mu_\ell, \theta)} = \frac{d_{s_{i-1}}(\mu)}{d(\mu_\ell, \theta)}$$

holds. If $L(s_{i-1}) = \text{'MIN'}$, $w_\ell^{s_0}(\mu) > 0$ implies

$$w_\ell^{s_{i-1}}(\mu) = \frac{w_\ell^{s_i}(\mu)/\tilde{d}_{s_i}(\mu)}{\sum_{c \in \mathcal{C}(s_{i-1})} 1/\tilde{d}_c(\mu)} \text{ and } \tilde{d}_{s_{i-1}}(\mu) = \frac{1}{\sum_{c \in \mathcal{C}(s_{i-1})} 1/\tilde{d}_c(\mu)},$$

thus

$$w_\ell^{s_{i-1}}(\mu) = \frac{w_\ell^{s_i}(\mu)/\tilde{d}_{s_i}(\mu)}{\sum_{c \in \mathcal{C}(s_{i-1})} 1/\tilde{d}_c(\mu)} = \frac{w_\ell^{s_i}(\mu)\tilde{d}_{s_{i-1}}(\mu)}{\tilde{d}_{s_i}(\mu)} = \frac{d_{s_i}(\mu)}{d(\mu_\ell, \theta)} \times \frac{|\tilde{d}_{s_{i-1}}(\mu)|}{|\tilde{d}_{s_i}(\mu)|} = \frac{d_{s_{i-1}}(\mu)}{d(\mu_\ell, \theta)}$$

holds. We can prove Eq. (12) similarly for the case with $a_{s_0}(\mu) = \text{'lose'}$. $\square$

Lemma C.5. On the event $\mathcal{E} = \left\{ \frac{N_\ell(t)}{t} \xrightarrow{t \rightarrow \infty} w_\ell^{s_0}(\mu) \text{ for all } \ell \in \mathcal{L}(\mathcal{T}) \right\}$,

$$\frac{Z_s(t)}{t} \xrightarrow{t \rightarrow \infty} \begin{cases} d_{s_0}(\mu) & (w_\ell^{s_0}(\mu) > 0 \text{ for some } \ell \in \mathcal{D}(s)) \\ 0 & (w_\ell^{s_0}(\mu) = 0 \text{ for all } \ell \in \mathcal{D}(s)) \end{cases} \quad (13)$$

holds for any node $s \in S$.

Proof. We prove the lemma by mathematical induction.

First, we prove (13) in the case with $s \in \mathcal{L}(\mathcal{T})$. For a leaf node s with $w_s^{s_0}(\mu) > 0$, $w_s^{s_0}(\mu) = d_{s_0}(\mu)/d(\mu_s, \theta)$ by Proposition C.4. By the assumption that $\frac{N_\ell(t)}{t} \xrightarrow{t \rightarrow \infty} w_\ell^{s_0}(\mu) > 0$, the leaf node s is selected infinitely often which leads to $\hat{\mu}_s(t) \xrightarrow{t \rightarrow \infty} \mu_s$ by law of large numbers. Thus,

$$\frac{Z_s(t)}{t} = \frac{N_s(t)d(\hat{\mu}_s(t), \theta)}{t} \xrightarrow{t \rightarrow \infty} w_s^{s_0}(\mu)d(\mu_s, \theta) = \frac{d_{s_0}(\mu)}{d(\mu_s, \theta)}d(\mu_s, \theta) = d_{s_0}(\mu)$$

converges to $w_s^{s_0}d(\mu_s, \theta) = d_{s_0}(\mu)$. As for a leaf node s with $w_s^{s_0}(\mu) = 0$, $N_s(t)/t \xrightarrow{t \rightarrow \infty} w_s^{s_0}(\mu) = 0$ by the assumption. Thus,

$$\frac{Z_s(t)}{t} = \frac{N_s(t)d(\hat{\mu}_s(t), \theta)}{t} \text{ or } \frac{0}{t} \xrightarrow{t \rightarrow \infty} 0 \times d(\mu_s, \theta) \text{ or } 0 = 0.$$

Next, we prove (13) for node $s \in S$, assuming that (13) holds for all the child nodes $c \in \mathcal{C}(s)$.

Consider the case with $(L(s) = \text{'MAX' and } a_{s_0}(\mu) = \text{'win'})$ or $(L(s) = \text{'MIN' and } a_{s_0}(\mu) = \text{'lose'})$.

For node s with $w_\ell^{s_0}(\mu) > 0$ for some $\ell \in \mathcal{D}(s)$

$$\frac{Z_s(t)}{t} = \frac{\sum_{c \in \mathcal{C}(s)} Z_c(t)}{t} = \frac{Z_{c^*(s)}(t)}{t} + \sum_{c \in \mathcal{C}(s) \setminus \{c^*(s)\}} \frac{Z_c(t)}{t} \xrightarrow{t \rightarrow \infty} d_{s_0}(\mu) + \sum_{c \in \mathcal{C}(s) \setminus \{c^*(s)\}} 0 = d_{s_0}(\mu)$$

by inductive assumption because $c \in \mathcal{C}(s)$ satisfies $w_\ell^{s_0}(\mu) > 0$ for some $\ell \in \mathcal{D}(c)$ if and only if $c = c^*(s)$, where $c^*(s) = c^+(s)$ if $L(s) = \text{'MAX'}$ and $c^*(s) = c^-(s)$ if $L(s) = \text{'MIN'}$.

553 For node s with $w_\ell^{s_0}(\mu) = 0$ for all $\ell \in \mathcal{D}(s)$,

$$\frac{Z_s(t)}{t} = \sum_{c \in \mathcal{C}(s)} \frac{Z_c(t)}{t} \xrightarrow{t \rightarrow \infty} \sum_{c \in \mathcal{C}(s)} 0 = 0$$

554 by inductive assumption.

555 Consider the case with $(L(s) = \text{'MIN'} \text{ and } a_{s_0}(\mu) = \text{'win'})$ or $(L(s) = \text{'MAX'} \text{ and } a_{s_0}(\mu) =$
 556 $\text{'lose'})$.

557 For node s with $w_\ell^{s_0}(\mu) > 0$ for some $\ell \in \mathcal{D}(s)$,

$$\frac{Z_s(t)}{t} = \min_{c \in \mathcal{C}(s)} \frac{Z_c(t)}{t} \xrightarrow{t \rightarrow \infty} \min_{c \in \mathcal{C}(s)} d_{s_0}(\mu) = d_{s_0}(\mu)$$

558 by inductive assumption.

559 For node s with $w_\ell^{s_0}(\mu) = 0$ for all $\ell \in \mathcal{D}(s)$,

$$\frac{Z_s(t)}{t} = \min_{c \in \mathcal{C}(s)} \frac{Z_c(t)}{t} \xrightarrow{t \rightarrow \infty} \min_{c \in \mathcal{C}(s)} 0 = 0$$

560 by inductive assumption.

561 Therefore, (13) holds for all $s \in S$. □

562 **Proposition C.6.** For any three constants $c_1, c_2, c_3 > 0$ with $2c_2c_3 \geq c_1$,

$$x = \frac{1}{c_1} \left[\ln c_2 + \ln \ln \frac{c_3}{c_1} + \ln \ln \ln \left(\frac{2c_2c_3}{c_1} \right)^2 \right]$$

563 satisfies $c_1x \geq \ln(c_2 \ln c_3x)$.

564 *Proof.* By substituting for x , we obtain

$$c_1x = \ln c_2 + \ln \ln \frac{c_3}{c_1} + \ln \ln \ln \left(\frac{2c_2c_3}{c_1} \right)^2$$

565 and

$$\begin{aligned} \ln(c_2 \ln c_3x) &= \ln c_2 + \ln \ln \left(\frac{c_3}{c_1} \left[\ln c_2 + \ln \ln \frac{c_3}{c_1} + \ln \ln \ln \left(\frac{2c_2c_3}{c_1} \right)^2 \right] \right) \\ &= \ln c_2 + \ln \ln \frac{c_3}{c_1} + \ln \ln \ln \left(c_2 \ln \left(\frac{c_3}{c_1} \ln \left(\frac{2c_2c_3}{c_1} \right)^2 \right) \right). \end{aligned}$$

566 Since

$$\begin{aligned} c_2 \ln \left(\frac{c_3}{c_1} \ln \left(\frac{2c_2c_3}{c_1} \right)^2 \right) &\leq c_2 \ln \left(\frac{2c_3}{c_1} \left(\frac{2c_2c_3}{c_1} - 1 \right) \right) \\ &\leq c_2 \left(\frac{2c_3}{c_1} \left(\frac{2c_2c_3}{c_1} - 1 \right) - 1 \right) \\ &< \left(\frac{2c_2c_3}{c_1} \right)^2, \end{aligned}$$

567 $c_1x \geq \ln(c_2 \ln c_3x)$ holds. □

568 **Proposition C.7.** For the function $\mathcal{C}_{\exp}(x)$ defined by Eq. (9),

$$\mathcal{C}_{\exp}(x) = x + o(x)$$

569 holds.

570 *Proof.* Let $X = \frac{h^{-1}(x+1)+\ln(\pi^2/3)}{2}$. Since inequality

$$h^{-1}(x) \leq x + \ln \left(x + \sqrt{2(x-1)} \right) \quad (14)$$

571 holds for $x \geq 1$ by Proposition 8 in [Kaufmann and Koolen, 2021],

$$X \leq \frac{x+1+\ln(1+x+\sqrt{2x})+\ln(\pi^2/3)}{2}$$

572 holds. Thus,

$$\begin{aligned} h^{-1}(X) &\leq X + \ln \left(X + \sqrt{2(X-1)} \right) \\ &\leq \frac{x+1+\ln(1+x+\sqrt{2x})+\ln(\pi^2/3)}{2} \\ &\quad + \ln \left(\frac{x+1+\ln(1+x+\sqrt{2x})+\ln(\pi^2/3)}{2} + \sqrt{x-1+\ln(1+x+\sqrt{2x})+\ln(\pi^2/3)} \right) \\ &= \frac{x}{2} + o(x) \end{aligned} \quad (15)$$

573 holds. For $X \geq h(1/\ln(3/2))$,

$$\begin{aligned} \mathcal{C}_{\exp}(x) &= 2\tilde{h}(X) \\ &= 2e^{1/h^{-1}(X)} h^{-1}(X) \\ &\leq 2 \left(1 + e \frac{1}{h^{-1}(X)} \right) h^{-1}(X) \\ &= 2h^{-1}(X) + 2e \end{aligned} \quad (16)$$

574 holds for $h^{-1}(X) \geq 1$. Therefore, $\mathcal{C}_{\exp}(x) = x + o(x)$ holds by Ineqs. (15) and (16). $\square$

575 **Lemma C.8.** Function $\beta(t, \delta)$ defined by Eq. (8) satisfies

$$\inf \{t \in \mathbb{N} : at \geq \beta(t, \delta)\} \leq \frac{1}{a} \ln \frac{1}{\delta} + o \left(\ln \frac{1}{\delta} \right)$$

576 for $a > 0$.

577 *Proof.* We have

$$\begin{aligned} &\inf \{t \in \mathbb{N} : at \geq \beta(t, \delta)\} \\ &= \inf \left\{ t \in \mathbb{N} : at \geq 3 \sum_{\ell \in \mathcal{L}(\mathcal{T})} \ln(1 + \ln N_{\ell}(t)) + |\mathcal{L}(\mathcal{T})| \mathcal{C}_{\exp} \left(\frac{\ln(1/\delta)}{|\mathcal{L}(\mathcal{T})|} \right) \right\} \\ &\quad \text{(by definition of } \beta(t, \delta) \text{)} \\ &\leq \inf \left\{ t \in \mathbb{N} : at \geq 3|\mathcal{L}(\mathcal{T})| \ln \left(1 + \ln \frac{t}{|\mathcal{L}(\mathcal{T})|} \right) + |\mathcal{L}(\mathcal{T})| \mathcal{C}_{\exp} \left(\frac{\ln(1/\delta)}{|\mathcal{L}(\mathcal{T})|} \right) \right\} \\ &\quad \text{(by Jensen's inequality for } f(x) = \ln(1 + \ln x) \text{ (} x \geq 1 \text{))} \\ &= \inf \left\{ t \in \mathbb{N} : \frac{at}{3|\mathcal{L}(\mathcal{T})|} \geq \ln \left(e^{\frac{1}{3}\mathcal{C}_{\exp}(\frac{\ln(1/\delta)}{|\mathcal{L}(\mathcal{T})|})} \ln \frac{et}{|\mathcal{L}(\mathcal{T})|} \right) \right\} \\ &\leq \frac{3|\mathcal{L}(\mathcal{T})|}{a} \left(\frac{1}{3}\mathcal{C}_{\exp} \left(\frac{\ln(1/\delta)}{|\mathcal{L}(\mathcal{T})|} \right) + \ln \ln \frac{3e}{a} + \ln \ln \ln \left(e^{\frac{1}{3}\mathcal{C}_{\exp}(\frac{\ln(1/\delta)}{|\mathcal{L}(\mathcal{T})|})} \frac{6e}{a} \right)^2 \right) \text{ (by Proposition C.6)} \\ &= \frac{3|\mathcal{L}(\mathcal{T})|}{a} \left(\frac{1}{3}\mathcal{C}_{\exp} \left(\frac{\ln(1/\delta)}{|\mathcal{L}(\mathcal{T})|} \right) + \ln \ln \frac{3e}{a} + \ln \ln \left(\frac{2}{3}\mathcal{C}_{\exp} \left(\frac{\ln(1/\delta)}{|\mathcal{L}(\mathcal{T})|} \right) + 2 \ln \frac{6e}{a} \right) \right) \\ &= \frac{1}{a} \log \frac{1}{\delta} + o \left(\frac{\ln(1/\delta)}{|\mathcal{L}(\mathcal{T})|} \right) + \frac{3|\mathcal{L}(\mathcal{T})|}{a} \left(\ln \ln \frac{3e}{a} + \ln \ln \left(\frac{2 \ln(1/\delta)}{3|\mathcal{L}(\mathcal{T})|} + o \left(\frac{\ln(1/\delta)}{|\mathcal{L}(\mathcal{T})|} \right) + 2 \ln \frac{6e}{a} \right) \right) \\ &\quad \text{(by Proposition C.7)} \\ &= \frac{1}{a} \ln \frac{1}{\delta} + o \left(\ln \frac{1}{\delta} \right). \end{aligned}$$

578 $\square$

579 Finally, we provide the proof of Lemma C.1.

580 *Proof of Lemma C.1.* For all $\epsilon > 0$, there exists t_1 such that $(1 + \epsilon)Z_{s_0}(t)/t \geq d_{s_0}(\boldsymbol{\mu})$ for all $t \geq t_1$
 581 by Lemma C.5. Thus,

$$\begin{aligned} \tau_\delta &= \inf \{t \in \mathbb{N} : Z_{s_0}(t) \geq \beta(t, \delta)\} \\ &\leq t_1 \vee \inf \{t \in \mathbb{N} : t(1 + \epsilon)^{-1}d_{s_0}(\boldsymbol{\mu}) \geq \beta(t, \delta)\} \\ &= t_1 \vee \frac{(1 + \epsilon) \ln(1/\delta)}{d_{s_0}(\boldsymbol{\mu})} + o(\ln(1/\delta)), \end{aligned} \quad (\text{by Lemma C.8})$$

582 where $a \vee b = \max\{a, b\}$ for any $a, b \in \mathbb{N}$. Therefore, $\mathbb{P}(\tau_\delta < +\infty) = 1$. Since t_1 does not depend
 583 on δ , we obtain the asymptotically upper bound

$$\limsup_{\delta \rightarrow +0} \frac{\tau_\delta}{\log(1/\delta)} \leq \frac{1 + \epsilon}{d_{s_0}(\boldsymbol{\mu})}.$$

584 Setting $\epsilon \rightarrow +0$, the proof is complete. $\square$

585 C.2 Continuity of the Optimal Proportion

586 Under Assumption 5.1 and 5.2, we show its continuity at $\boldsymbol{\mu}$ for the Thresholding MCTS problem.

587 **Lemma C.9** (Continuity of the optimal proportion $w^{s_0}(\cdot)$ at $\boldsymbol{\mu}$). *For any $\epsilon > 0$, there exists $0 < \xi$*
 588 *such that*

$$\max_{\ell \in \mathcal{L}(\mathcal{T})} |w_\ell^{s_0}(\boldsymbol{\mu}') - w_\ell^{s_0}(\boldsymbol{\mu})| < \epsilon \text{ for all } \boldsymbol{\mu}' \in \prod_{\ell \in \mathcal{L}(\mathcal{T})} [\mu_\ell - \xi, \mu_\ell + \xi].$$

589 The function $w_\ell^{s_0}(\cdot)$ is continuous not in the whole mean-vector domain but in a part of the domain
 590 that includes $\boldsymbol{\mu}$. The following propositions, a corollary, and a lemma are used to make clear such a
 591 part of the domain.

592 The first proposition states the continuity of the function $V_s(\cdot)$ for $s \in S$.

593 **Proposition C.10.** $|\mu'_\ell - \mu_\ell| < \epsilon$ for $\ell \in \mathcal{L}(\mathcal{T}) \Rightarrow |V_s(\boldsymbol{\mu}') - V_s(\boldsymbol{\mu})| < \epsilon$ for $s \in S$.

594 *Proof.* This can be proved by mathematical induction. For $s \in \mathcal{L}(\mathcal{T})$, $|V_s(\boldsymbol{\mu}') - V_s(\boldsymbol{\mu})| < \epsilon$ holds
 595 by the assumption.

596 Assume that $|V_c(\boldsymbol{\mu}') - V_c(\boldsymbol{\mu})| < \epsilon$ for all $c \in \mathcal{C}(s)$ for node $s \in S \setminus \mathcal{L}(\mathcal{T})$. Since $V_c(\boldsymbol{\mu}) - \epsilon <$
 597 $V_c(\boldsymbol{\mu}') < V_c(\boldsymbol{\mu}) + \epsilon$,

$$\begin{aligned} \max_{c \in \mathcal{C}(s)} V_c(\boldsymbol{\mu}) - \epsilon &< \max_{c \in \mathcal{C}(s)} V_c(\boldsymbol{\mu}') < \max_{c \in \mathcal{C}(s)} V_c(\boldsymbol{\mu}) + \epsilon \text{ and} \\ \min_{c \in \mathcal{C}(s)} V_c(\boldsymbol{\mu}) - \epsilon &< \min_{c \in \mathcal{C}(s)} V_c(\boldsymbol{\mu}') < \min_{c \in \mathcal{C}(s)} V_c(\boldsymbol{\mu}) + \epsilon \end{aligned}$$

598 hold. For node s , $V_s(\boldsymbol{\mu}) = \max_{c \in \mathcal{C}(s)} V_c(\boldsymbol{\mu})$ if $L(s) = \text{'MAX'}$ and $V_s(\boldsymbol{\mu}) = \min_{c \in \mathcal{C}(s)} V_c(\boldsymbol{\mu})$ if
 599 $L(s) = \text{'MIN'}$, thus in both cases, $V_s(\boldsymbol{\mu}) - \epsilon < V_s(\boldsymbol{\mu}') < V_s(\boldsymbol{\mu}) + \epsilon$ holds. Therefore, $|V_s(\boldsymbol{\mu}') -$
 600 $V_s(\boldsymbol{\mu})| < \epsilon$ holds. $\square$

601 We have the following corollary of this proposition.

602 **Corollary C.11.** $|\mu'_\ell - \mu_\ell| < |V_{s_0}(\boldsymbol{\mu}) - \theta|$ for $\ell \in \mathcal{L}(\mathcal{T}) \Rightarrow a_{s_0}(\boldsymbol{\mu}') = a_{s_0}(\boldsymbol{\mu})$

603 *Proof.* By proposition C.10, $|\mu'_\ell - \mu_\ell| < |V_{s_0}(\boldsymbol{\mu}) - \theta|$ for $\ell \in \mathcal{L}(\mathcal{T})$ implies $|V_{s_0}(\boldsymbol{\mu}') - V_{s_0}(\boldsymbol{\mu})| <$
 604 $|V_{s_0}(\boldsymbol{\mu}) - \theta|$. Thus,

$$V_{s_0}(\boldsymbol{\mu}) - |V_{s_0}(\boldsymbol{\mu}) - \theta| < V_{s_0}(\boldsymbol{\mu}') < V_{s_0}(\boldsymbol{\mu}) + |V_{s_0}(\boldsymbol{\mu}) - \theta|$$

605 Thus,

$$\begin{aligned} V_{s_0}(\boldsymbol{\mu}) \geq \theta &\Rightarrow V_{s_0}(\boldsymbol{\mu}') > V_{s_0}(\boldsymbol{\mu}) - (V_{s_0}(\boldsymbol{\mu}) - \theta) = \theta \text{ and} \\ V_{s_0}(\boldsymbol{\mu}) < \theta &\Rightarrow V_{s_0}(\boldsymbol{\mu}') < V_{s_0}(\boldsymbol{\mu}) + (\theta - V_{s_0}(\boldsymbol{\mu})) = \theta \end{aligned}$$

606 hold, which means $a_{s_0}(\boldsymbol{\mu}') = a_{s_0}(\boldsymbol{\mu})$. $\square$

607 The next proposition is used to prove Proposition C.13.

608 **Proposition C.12.** For any positive real numbers $a_1, a_2, \dots, a_n$ with $n \geq 1$, the following inequalities hold.
 609

610 (1) For $0 < \epsilon < \min_{i=1,2,\dots,n} a_i$,

$$\frac{1}{\sum_{i=1}^n \frac{1}{a_i}} - \epsilon < \frac{1}{\sum_{i=1}^n \frac{1}{a_i - \epsilon}}.$$

611 (2) For $\epsilon > 0$,

$$\frac{1}{\sum_{i=1}^n \frac{1}{a_i + \epsilon}} < \frac{1}{\sum_{i=1}^n \frac{1}{a_i}} + \epsilon.$$

612 *Proof.* Define $f(x)$ as

$$f(x) = \frac{1}{\sum_{i=1}^n \frac{1}{a_i}} + x - \frac{1}{\sum_{i=1}^n \frac{1}{a_i + x}}.$$

613 Then, $f(x)$ is continuous on $(-\min_{i=1,2,\dots,n} a_i, \infty)$. Since

$$\frac{d}{dx} f(x) = 1 - \frac{\sum_{i=1}^n \frac{1}{(a_i + x)^2}}{\left(\sum_{i=1}^n \frac{1}{a_i + x}\right)^2} > 0$$

614 on $(-\min_{i=1,2,\dots,n} a_i, \infty)$ and $f(0) = 0$, (1) $f(x) < 0$ on $(\min_{i=1,2,\dots,n} a_i, 0)$ and (2) $f(x) > 0$
 615 on $(0, \infty)$ hold. $\square$

616 The following proposition is on the continuity of function $d_s(\cdot)$ for $s \in S$.

617 **Proposition C.13.** $|d_\ell(\mu') - d_\ell(\mu)| < \epsilon$ for $\ell \in \mathcal{L}(\mathcal{T}) \Rightarrow |d_s(\mu') - d_s(\mu)| < \epsilon$ for $s \in S$.

618 *Proof.* This can be proved by mathematical induction. For $s \in \mathcal{L}(\mathcal{T})$, $|d_s(\mu') - d_s(\mu)| < \epsilon$ holds
 619 by the assumption.

620 Assume that $|d_c(\mu') - d_c(\mu)| < \epsilon$ for all $c \in \mathcal{C}(s)$ for node $s \in S \setminus \mathcal{L}(\mathcal{T})$, and we obtain
 621 $d_c(\mu) - \epsilon < d_c(\mu') < d_c(\mu) + \epsilon$.

622 First, we consider $a_{s_0}(\mu') = \text{'win'}$.

623 In the case with $L(s) = \text{'MAX'}$,

$$\max_{c \in \mathcal{C}(s)} d_c(\mu) - \epsilon < \max_{c \in \mathcal{C}(s)} d_c(\mu') < \max_{c \in \mathcal{C}(s)} d_c(\mu) + \epsilon$$

624 holds. Since $d_s(\mu) = \max_{c \in \mathcal{C}(s)} d_c(\mu)$,

$$d_s(\mu) - \epsilon < d_s(\mu') < d_s(\mu) + \epsilon$$

625 holds.

626 Consider the case with $L(s) = \text{'MIN'}$.

627 If $d_c(\mu) = d_{c'}(\mu') = 0$ for some $c, c' \in \mathcal{C}(s)$, then $d_s(\mu) = d_s(\mu') = 0$. Thus, $|d_s(\mu') - d_s(\mu)| < \epsilon$
 628 holds.

629 If $d_c(\mu) = 0$ for some $c \in \mathcal{C}(s)$ and $d_c(\mu') > 0$ for all $c \in \mathcal{C}(s)$, then $d_s(\mu) = 0$ and

$$d_s(\mu') = \frac{1}{\sum_{c \in \mathcal{C}(s)} \frac{1}{d_c(\mu')}} < \frac{1}{\frac{1}{\epsilon}} = \epsilon.$$

630 Thus, $|d_s(\mu') - d_s(\mu)| < \epsilon$ holds.

631 If $d_c(\mu) > 0$ for all $c \in \mathcal{C}(s)$ and $d_c(\mu') = 0$ for some $c \in \mathcal{C}(s)$, then $d_c(\mu) < \epsilon$ by the induction
 632 assumption. In this case, $d_s(\mu') = 0$ and

$$d_s(\mu) = \frac{1}{\sum_{c \in \mathcal{C}(s)} \frac{1}{d_c(\mu)}} < \frac{1}{\frac{1}{\epsilon}} = \epsilon.$$

633 Thus, $|d_s(\mu') - d_s(\mu)| < \epsilon$ holds.

634 If $d_c(\mu), d_c(\mu') > 0$ for all $c \in \mathcal{C}(s)$, then

$$d_s(\mu') = \frac{1}{\sum_{c \in \mathcal{C}(s)} \frac{1}{d_c(\mu')}} < \frac{1}{\sum_{c \in \mathcal{C}(s)} \frac{1}{d_c(\mu) + \epsilon}} \leq \frac{1}{\sum_{c \in \mathcal{C}(s)} \frac{1}{d_c(\mu)}} + \epsilon = d_s(\mu) + \epsilon$$

635 holds, where the last inequality holds by Proposition C.12. When $d_c(\mu) > \epsilon$ for all $c \in \mathcal{C}(s)$,

$$d_s(\mu) - \epsilon = \frac{1}{\sum_{c \in \mathcal{C}(s)} \frac{1}{d_c(\mu)}} - \epsilon \leq \frac{1}{\sum_{c \in \mathcal{C}(s)} \frac{1}{d_c(\mu) - \epsilon}} < \frac{1}{\sum_{c \in \mathcal{C}(s)} \frac{1}{d_c(\mu')}} = d_s(\mu')$$

636 holds, where the first inequality holds by Proposition C.12. when $d_c(\mu) \leq \epsilon$ for some $c \in \mathcal{C}(s)$,

$$d_s(\mu) - \epsilon = \frac{1}{\sum_{c \in \mathcal{C}(s)} \frac{1}{d_c(\mu)}} - \epsilon \leq \frac{1}{\frac{1}{\epsilon}} - \epsilon = 0 < d_s(\mu')$$

637 holds. Therefore, $|d_s(\mu') - d_s(\mu)| < \epsilon$ holds.

638 Similarly, we can show $|d_s(\mu') - d_s(\mu)| < \epsilon$ even if $a_{s_0}(\mu') = \text{'lose'}$. $\square$

639 The following proposition states a relation between the functions $d_s(\cdot)$ and $V_s(\cdot)$, and is used in the
640 proof of Proposition C.15.

641 **Proposition C.14.** For $s \in S$,

$$d_s(\mu) > 0 \Leftrightarrow V_s(\mu) \begin{cases} > \theta & (a_{s_0}(\mu) = \text{'win'}) \\ < \theta & (a_{s_0}(\mu) = \text{'lose'}). \end{cases}$$

642 *Proof.* For $s \in \mathcal{L}(\mathcal{T})$,

$$\begin{cases} d_s(\mu) = d(\mu_s, \theta) > 0 \Leftrightarrow V_s(\mu) = \mu_s > \theta & (a_{s_0}(\mu) = \text{'win'}) \\ d_s(\mu) = d(\mu_s, \theta) > 0 \Leftrightarrow V_s(\mu) = \mu_s < \theta & (a_{s_0}(\mu) = \text{'lose'}) \end{cases}$$

643 hold. Thus, proposition statement holds in this case.

644 Assume that the proposition statement holds for $s \in \mathcal{C}(s')$.

645 In the case with $L(s') = \text{'MAX'}$,

$$\begin{cases} d_{s'}(\mu) = \max_{s \in \mathcal{C}(s')} d_s(\mu) > 0 \Leftrightarrow V_{s'}(\mu) = \max_{s \in \mathcal{C}(s')} V_s(\mu) > \theta & (a_{s_0}(\mu) = \text{'win'}) \\ d_{s'}(\mu) > 0 \Leftrightarrow d_s(\mu) > 0 \text{ for } s \in \mathcal{C}(s') \Leftrightarrow V_{s'}(\mu) = \max_{s \in \mathcal{C}(s')} V_s(\mu) < \theta & (a_{s_0}(\mu) = \text{'lose'}). \end{cases}$$

646 In the case with $L(s') = \text{'MIN'}$,

$$\begin{cases} d_{s'}(\mu) > 0 \Leftrightarrow d_s(\mu) > 0 \text{ for } s \in \mathcal{C}(s') \Leftrightarrow V_{s'}(\mu) = \min_{s \in \mathcal{C}(s')} V_s(\mu) > \theta & (a_{s_0}(\mu) = \text{'win'}) \\ d_{s'}(\mu) = \max_{s \in \mathcal{C}(s')} d_s(\mu) > 0 \Leftrightarrow V_{s'}(\mu) = \min_{s \in \mathcal{C}(s')} V_s(\mu) < \theta & (a_{s_0}(\mu) = \text{'lose'}). \end{cases}$$

647 Thus, the proposition statement holds for s' . Therefore, the proposition statement holds for $s \in S$ by
648 induction. $\square$

649 The next proposition makes clear the necessary and sufficient condition of $w_\ell^{s_0}(\mu) > 0$ for leaf nodes
650 $\ell \in \mathcal{L}(\mathcal{T})$, and is used in the proof of Lemma C.16.

651 **Proposition C.15.** Let $s_0, s_1, \dots, s_k = \ell$ denote the path from node s_0 to ℓ in $\mathcal{T}$. Then,

$$w_\ell^{s_0}(\mu) > 0 \Leftrightarrow d_{s_i}(\mu) > 0 \text{ for } i = 0, 1, \dots, k \text{ and}$$

$$s_{i+1} = \begin{cases} c^+(s_i) \\ c^-(s_i) \end{cases} \text{ for } i = 0, 1, \dots, k-1 \text{ with } L(s_i) = \begin{cases} \text{'MAX'} & (a_{s_0}(\mu) = \text{'win'}) \\ \text{'MIN'} & (a_{s_0}(\mu) = \text{'lose'}). \end{cases}$$

652 *Proof.* We show the case with $a_{s_0}(\mu) = \text{'win'}$. Similar proof works for the case with $a_{s_0}(\mu) =$
653 'lose' .
654 $(\Rightarrow)$ By Assumption 5.1, $V_{s_0}(\mu) > \theta$ in this case, which implies $d_{s_0}(\mu) > 0$ by Proposition C.14.
655 Assume $w_\ell^{s_0}(\mu) > 0$. We prove $d_{s_i}(\mu) > 0$ for all $i = 0, 1, \dots, k$ by contradiction. Assume
656 $d_{s_i}(\mu) = 0$ for some $i \in \{0, 1, \dots, k\}$. Let i_0 be the smallest such index, that is, $d_{s_{i_0}}(\mu) > 0$ for
657 $i = 0, 1, \dots, i_0 - 1$ and $d_{s_{i_0}}(\mu) = 0$. Since $d_{s_0}(\mu) > 0$, $i_0 > 0$. $L(s_{i_0-1}) \neq \text{'MIN'}$ because
658 $d_{s_{i_0-1}}(\mu) > 0$ and $d_{s_{i_0}}(\mu) = 0$. Thus, $L(s_{i_0-1}) = \text{'MAX'}$ and $s_{i_0} \neq c^+(s_{i_0-1})$. Therefore,
659 $w_\ell^{s_{i_0-1}}(\mu) = 0$. The fact that $d_{s_i}(\mu) > 0$ for $i = 0, 1, \dots, i_0 - 1$ implies $L(s_i) = \text{'MAX'}$ or
660 $(L(s_i) = \text{'MIN'}$ and $d_c(\mu) > 0$ for $c \in \mathcal{C}(s_i)$) for $i = 0, 1, \dots, i_0 - 1$. Thus, $w_\ell^{s_i}(\mu) = w_\ell^{s_{i+1}}(\mu)$ or
661 $w_\ell^{s_i}(\mu) = \frac{w_\ell^{s_{i+1}}(\mu)/d_{s_{i+1}}(\mu)}{\sum_{c \in \mathcal{C}(s_i)} 1/d_c(\mu)}$ for $i = 0, 1, \dots, i_0 - 1$. Hence, $w_\ell^{s_0} = 0$, which contradicts the assump-
662 tion. Therefore, $d_{s_i}(\mu) > 0$ for all $i = 0, 1, \dots, k$. In this case, $s_{i+1} = c^+(s_i)$ for $i = 0, 1, \dots, k - 1$
663 with $L(s_i) = \text{'MAX'}$ because, if not, $w_\ell^{s_0}(\mu) = 0$ by a similar reason, which also contradicts the
664 assumption.
665 $(\Leftarrow)$ Assume that $d_{s_i}(\mu) > 0$ for $i = 0, 1, \dots, k$ and $s_{i+1} = c^+(s_i)$ for $i = 0, 1, \dots, k - 1$
666 with $L(s_i) = \text{'MAX'}$. Then, $L(s_i) = \text{'MAX'}$ or $(L(s_i) = \text{'MIN'}$ and $d_c(\mu) > 0$ for $c \in \mathcal{C}(s_i)$)
667 for $i = 0, 1, \dots, k - 1$. Therefore, $w_\ell^{s_i}(\mu) = w_\ell^{s_{i+1}}(\mu)$ or $w_\ell^{s_i}(\mu) = \frac{w_\ell^{s_{i+1}}(\mu)/d_{s_{i+1}}(\mu)}{\sum_{c \in \mathcal{C}(s_i)} 1/d_c(\mu)}$ for
668 $i = 0, 1, \dots, k - 1$. Since $w_\ell^{s_k}(\mu) > 0$, $w_\ell^{s_0}(\mu) > 0$. $\square$

669 The following lemma shows the part of the domain that guarantees $w^{s_0}(\cdot)$ has the same non-zero
670 components.

671 **Lemma C.16.** *Let*

$$\begin{aligned} \mathcal{L}(\mathcal{T})_\mu &= \{\ell \in \mathcal{L}(\mathcal{T}) \mid w_\ell^{s_0}(\mu) > 0\} \\ c_\mu^+(s) &= \arg \max_{c \in \mathcal{C}(s)} \tilde{d}_c(\mu) \\ c_\mu^-(s) &= \arg \min_{c \in \mathcal{C}(s)} \tilde{d}_c(\mu) \\ \Gamma_\mu^{\text{abs}} &= \min_{s \in S: \mathcal{D}(s) \cap \mathcal{L}(\mathcal{T})_\mu \neq \emptyset} d_s(\mu) \\ \Gamma_\mu^* &= \begin{cases} \min_{s \in S: \mathcal{D}(s) \cap \mathcal{L}(\mathcal{T})_\mu \neq \emptyset, L(s) = \text{'MAX'}} \min_{c \in \mathcal{C}(s) \setminus \{c_\mu^+(s)\}} \left(d_{c_\mu^+(s)}(\mu) - d_c(\mu) \right) & (a_{s_0}(\mu) = \text{'win'}) \\ \min_{s \in S: \mathcal{D}(s) \cap \mathcal{L}(\mathcal{T})_\mu \neq \emptyset, L(s) = \text{'MIN'}} \min_{c \in \mathcal{C}(s) \setminus \{c_\mu^-(s)\}} \left(d_{c_\mu^-(s)}(\mu) - d_c(\mu) \right) & (a_{s_0}(\mu) = \text{'lose'}). \end{cases} \end{aligned}$$

672 *Then,*

$$\begin{aligned} |\mu'_\ell - \mu_\ell| &< |V_{s_0}(\mu) - \theta|, |d_\ell(\mu') - d_\ell(\mu)| < \min\{\Gamma_\mu^*/2, \Gamma_\mu^{\text{abs}}\} \text{ for } \ell \in \mathcal{L}(\mathcal{T}) \\ \Rightarrow (w_\ell^{s_0}(\mu') > 0 \Leftrightarrow w_\ell^{s_0}(\mu) > 0) &\text{ for } \ell \in \mathcal{L}(\mathcal{T}). \end{aligned}$$

673 *Proof.* Assume that $|\mu'_\ell - \mu_\ell| < |V_{s_0}(\mu) - \theta|, |d_\ell(\mu') - d_\ell(\mu)| < \min\{\Gamma_\mu^*/2, \Gamma_\mu^{\text{abs}}\}$ for $\ell \in \mathcal{L}(\mathcal{T})$.
674 Since $|\mu'_\ell - \mu_\ell| < |V_{s_0}(\mu) - \theta|$ for $\ell \in \mathcal{L}(\mathcal{T})$, $a_{s_0}(\mu') = a_{s_0}(\mu)$ holds by Corollary C.11. In the
675 following, we prove the corollary inequality in the case with $a_{s_0}(\mu) = \text{'win'}$. We can prove the
676 inequality in the case with $a_{s_0}(\mu) = \text{'lose'}$ similarly. By proposition C.13, $|d_\ell(\mu') - d_\ell(\mu)| <$
677 $\min\{\Gamma_\mu^*/2, \Gamma_\mu^{\text{abs}}\}$ implies $|d_s(\mu') - d_s(\mu)| < \min\{\Gamma_\mu^*/2, \Gamma_\mu^{\text{abs}}\}$ for $s \in S$. For $\ell \in \mathcal{L}(\mathcal{T})$ with
678 $w_\ell^{s_0}(\mu) > 0$, let $s_0, s_1, \dots, s_k = \ell$ be the node sequence on the path from the root node s_0 to the
679 leaf node ℓ . Then, $d_{s_i}(\mu') > d_{s_i}(\mu) - \Gamma_\mu^{\text{abs}} \geq 0$ for $i = 0, 1, \dots, k$ and $c_{\mu'}^+(s_i) = c_\mu^+(s_i)$ for
680 $i = 0, 1, \dots, k - 1$ with $L(s_i) = \text{'MAX'}$ because $d_{c_{\mu'}^+(s_i)}(\mu') > d_{c_\mu^+(s_i)}(\mu) - \Gamma_\mu^*/2 \geq d_c(\mu) +$
681 $\Gamma_\mu^*/2 > d_c(\mu')$ for $c \in \mathcal{C}(s) \setminus \{c_\mu^+(s_i)\}$. Thus, $w_\ell^{s_0}(\mu') > 0$ by Proposition C.15. For $\ell \in \mathcal{L}(\mathcal{T})$
682 with $w_\ell^{s_0}(\mu) = 0$, let $i_0 = \min\{i \in \{0, 1, \dots, k\} \mid \mathcal{D}(s_i) \cap \mathcal{L}(\mathcal{T})_\mu \neq \emptyset\}$. Then, $L(s_{i_0-1}) = \text{'MAX'}$
683 and $s_{i_0} \neq c_\mu^+(s_{i_0-1})$. Thus, $d_{c_\mu^+(s_{i_0-1})}(\mu') > d_{c_\mu^+(s_{i_0-1})}(\mu) - \Gamma_\mu^*/2 \geq d_{s_{i_0}}(\mu) + \Gamma_\mu^*/2 > d_{s_{i_0}}(\mu')$
684 holds, which implies $w_\ell^{s_0}(\mu') = 0$. Therefore, $(w_\ell^{s_0}(\mu') > 0 \Leftrightarrow w_\ell^{s_0}(\mu) > 0)$ holds. $\square$

685 *Proof of Lemma C.9.* We prove the lemma in the case with $a_{s_0}(\mu) = \text{'win'}$. We can prove similarly
686 in the case with $a_{s_0}(\mu) = \text{'lose'}$.

687 Since Function

$$d_\ell(\boldsymbol{\mu}) = \begin{cases} d(\mu_\ell, \theta) & (\mu_\ell \geq \theta) \\ 0 & (\mu_\ell < 0) \end{cases}$$

688 is continuous, there exists $\xi_\mu > 0$ such that

$$\boldsymbol{\mu}' \in \prod_{\ell \in \mathcal{L}(\mathcal{T})} [\mu - \xi_\mu, \mu + \xi_\mu] \Rightarrow |\mu'_\ell - \mu_\ell| < |V_{s_0}(\boldsymbol{\mu}) - \theta|, |d_\ell(\boldsymbol{\mu}') - d_\ell(\boldsymbol{\mu})| < \min\{\Gamma_\mu^*/2, \Gamma_\mu^{\text{abs}}\} \text{ for } \ell \in \mathcal{L}(\mathcal{T}).$$

689 Then, $a_{s_0}(\boldsymbol{\mu}') = a_{s_0}(\boldsymbol{\mu})$ by Corollary C.11 and $(w_\ell^{s_0}(\boldsymbol{\mu}') > 0 \Leftrightarrow w_\ell^{s_0}(\boldsymbol{\mu}) > 0)$ for all $\ell \in \mathcal{L}(\mathcal{T})$ by
690 Lemma C.16. By Proposition C.4,

$$w_\ell^{s_0}(\boldsymbol{\mu}) = \frac{d_{s_0}(\boldsymbol{\mu})}{d(\mu_\ell, \theta)}, \quad w_\ell^{s_0}(\boldsymbol{\mu}') = \frac{d_{s_0}(\boldsymbol{\mu}')}{d(\mu'_\ell, \theta)}$$

691 for $\ell \in \mathcal{L}(\mathcal{T})$ with $w_\ell^{s_0}(\boldsymbol{\mu}) > 0$. Since $d(\mu_\ell, \theta) = d_\ell(\cdot)$ is continuous as above, and $d_{s_0}(\cdot)$ is
692 continuous by Proposition C.13, $w_\ell^{s_0}(\cdot)$ is also continuous. Thus, for any $\epsilon > 0$, there exists
693 $0 < \xi < \xi_\mu$ such that

$$\boldsymbol{\mu}' \in \prod_{\ell \in \mathcal{L}(\mathcal{T})} [\mu - \xi, \mu + \xi] \Rightarrow |w_\ell^{s_0}(\boldsymbol{\mu}') - w_\ell^{s_0}(\boldsymbol{\mu})| < \epsilon \text{ for } \ell \in \mathcal{L}(\mathcal{T}).$$

694

□

695 C.3 Proof of Lemma C.2

696 The following lemma is the RD-Tracking-TMCTS version of Lemma 17 of [Garivier and Kaufmann,
697 2016], which is used to prove Lemma C.2.

698 **Lemma C.17.** *For $\epsilon > 0$, assume that there exists a positive integer t_0 such that*

$$|w_\ell^{s_0}(\hat{\boldsymbol{\mu}}(t)) - w_\ell^{s_0}(\boldsymbol{\mu})| \leq \epsilon \text{ for all } t \geq t_0 \text{ and } \ell \in \mathcal{L}(\mathcal{T}). \quad (17)$$

699 *Then, the sampling rule of RD-Tracking-TMCTS ensures that there exists a positive integer $t_1 \geq t_0$
700 such that*

$$\max_{\ell \in \mathcal{L}(\mathcal{T})} \left| \frac{N_\ell(t)}{t} - w_\ell^{s_0}(\boldsymbol{\mu}) \right| \leq 2(|\mathcal{L}(\mathcal{T})| - 1)\epsilon \text{ for } t \geq t_1. \quad (18)$$

701 *Proof.* For all $\ell \in \mathcal{L}(\mathcal{T})$, let $E_{\ell,t} = N_\ell(t) - tw_\ell^{s_0}(\boldsymbol{\mu})$. Then

$$\max_{\ell \in \mathcal{L}(\mathcal{T})} |E_{\ell,t}| \leq (|\mathcal{L}(\mathcal{T})| - 1) \max_{\ell \in \mathcal{L}(\mathcal{T})} E_{\ell,t} \quad (19)$$

702 holds because $\sum_{\ell \in \mathcal{L}(\mathcal{T})} E_{\ell,t} = 0$.

703 Let $t'_0 \geq \max\{t_0, 1/\epsilon\}$ be a positive integer such that

$$\sqrt{t+1} - |\mathcal{L}(\mathcal{T})|/2 \leq t\epsilon \text{ for all } t \geq t'_0.$$

704 We can prove that

$$(I(t+1) = \ell) \subseteq (E_{\ell,t} \leq t\epsilon) \text{ for } t \geq t'_0 \quad (20)$$

705 as follows. By the sampling rule of RD-Tracking-TMCTS, we have

$$(I(t+1) = \ell) \subseteq \left(N_\ell(t) < \sqrt{t+1} - |\mathcal{L}(\mathcal{T})|/2 \text{ or } \ell = \arg \max_{\ell' \in \mathcal{L}(\mathcal{T})} \frac{w_{\ell'}^{s_0}(\hat{\boldsymbol{\mu}}(t))}{N_{\ell'}(t)} \right).$$

706 In the case with $N_\ell(t) < \sqrt{t+1} - |\mathcal{L}(\mathcal{T})|/2$,

$$E_{\ell,t} < \sqrt{t+1} - |\mathcal{L}(\mathcal{T})|/2 - tw_\ell^{s_0}(\boldsymbol{\mu}) \leq \sqrt{t+1} - |\mathcal{L}(\mathcal{T})|/2 \leq t\epsilon$$

707 for $t \geq t'_0$. In the case with $\ell = \arg \max_{\ell' \in \mathcal{L}(\mathcal{T})} \frac{w_{\ell'}^{s_0}(\hat{\boldsymbol{\mu}}(t))}{N_{\ell'}(t)}$,

$$\frac{w_\ell^{s_0}(\hat{\boldsymbol{\mu}}(t))}{\frac{N_\ell(t)}{t}} \geq 1$$

708 holds because $\mathbf{w}^{s_0}(\hat{\boldsymbol{\mu}}(t)), \left(\frac{N_{\ell'}(t)}{t}\right)_{\ell' \in \mathcal{L}(\mathcal{T})} \in \Delta$. Thus, we have

$$\begin{aligned} N_{\ell}(t) - t w_{\ell}^{s_0}(\hat{\boldsymbol{\mu}}(t)) &\leq 0 \\ N_{\ell}(t) - t w_{\ell}^{s_0}(\boldsymbol{\mu}) + t(w_{\ell}^{s_0}(\boldsymbol{\mu}) - w_{\ell}^{s_0}(\hat{\boldsymbol{\mu}}(t))) &\leq 0 \\ E_{\ell,t} &\leq -t(w_{\ell}^{s_0}(\boldsymbol{\mu}) - w_{\ell}^{s_0}(\hat{\boldsymbol{\mu}}(t))) \leq t\epsilon \end{aligned}$$

709 for $t \geq t'_0 \geq t_0$. Therefore,

$$\left(N_{\ell}(t) < \sqrt{t+1} - |\mathcal{L}(\mathcal{T})|/2 \text{ or } \ell = \arg \max_{\ell' \in \mathcal{L}(\mathcal{T})} \frac{w_{\ell'}^{s_0}(\hat{\boldsymbol{\mu}}(t))}{N_{\ell'}(t)} \right) \subseteq (E_{\ell,t} \leq t\epsilon)$$

710 holds for $t \geq t'_0$. Thus, (20) holds.

711 Since $E_{\ell,t+1} = E_{\ell,t} + \mathbb{1}\{I(t+1) = \ell\} - \mu_{\ell}$, (20) implies

$$E_{\ell,t+1} \leq E_{\ell,t} + \mathbb{1}\{E_{\ell,t} \leq t\epsilon\} - \mu_{\ell} \text{ for } t \geq t'_0.$$

712 Then, we can prove that

$$E_{\ell,t} \leq \max\{E_{\ell,t'_0}, t\epsilon + 1\} \quad (21)$$

713 for $t \geq t'_0$ by induction on t as follows. For $t = t'_0$, Ineq. (21) holds trivially. Assume that Ineq. (21)

714 holds for some $t \geq t'_0$. Then,

$$E_{\ell,t+1} \leq \begin{cases} t\epsilon + 1 - \mu_{\ell} \leq t\epsilon + 1 & (E_{\ell,t} \leq t\epsilon) \\ \max\{E_{\ell,t'_0}, t\epsilon + 1\} - \mu_{\ell} \leq \max\{E_{\ell,t'_0}, t\epsilon + 1\} & (E_{\ell,t} > t\epsilon). \end{cases}$$

715 Thus, $E_{\ell,t+1} \leq \max\{E_{\ell,t'_0}, (t+1)\epsilon + 1\}$ holds, which means that (21) holds for $t+1$. Therefore,

716 (21) holds for all $t \geq t'_0$.

717 For $t \geq t'_0$, by Ineq. (19) and Ineq. (21),

$$\max_{\ell \in \mathcal{L}(\mathcal{T})} \left| \frac{E_{\ell,t}}{t} \right| \leq (|\mathcal{L}(\mathcal{T})| - 1) \max \left\{ \frac{E_{\ell,t'_0}}{t}, \epsilon + \frac{1}{t} \right\} \leq (|\mathcal{L}(\mathcal{T})| - 1) \max \left\{ \frac{t'_0}{t}, 2\epsilon \right\}$$

718 holds, where the last inequality uses $E_{\ell,t'_0} \leq t'_0$ and $1/t \leq \epsilon$. Therefore, there exists $t_1 \geq t'_0$ such that

$$\max_{\ell \in \mathcal{L}(\mathcal{T})} \left| \frac{E_{\ell,t}}{t} \right| \leq 2(|\mathcal{L}(\mathcal{T})| - 1)\epsilon \text{ for } t \geq t_1,$$

719 which means this lemma's statement holds.

720 □

721 *Remark C.18.* If Assumption (17) is restricted as

$$|w_{\ell}^{s_0}(\hat{\boldsymbol{\mu}}(t)) - w_{\ell}^{s_0}(\boldsymbol{\mu})| \leq \epsilon \text{ for all } t_0 \leq t \leq T \text{ and } \ell \in \mathcal{L}(\mathcal{T}), \quad (22)$$

722 the existence $t_1 \geq t_0$ that satisfies

$$\max_{\ell \in \mathcal{L}(\mathcal{T})} \left| \frac{N_{\ell}(t)}{t} - w_{\ell}^{s_0}(\boldsymbol{\mu}) \right| \leq 2(|\mathcal{L}(\mathcal{T})| - 1)\epsilon \text{ for } t_1 \leq t \leq T \quad (23)$$

723 is still guaranteed.

724 *Proof of Lemma C.2.* Let $\epsilon > 0$ be an arbitrary positive real number. The forced exploration ($I(t) \leftarrow$
725 $\min_{\ell \in \mathcal{L}(\mathcal{T})} N_{\ell}(t-1)$ if $N_{\ell}(t-1) < \sqrt{t} - |\mathcal{L}(\mathcal{T})|/2$ for some $\ell \in \mathcal{L}(\mathcal{T})$) of RD-Tracking-TMCTS
726 sampling rule ensures $N_{\ell}(t) \rightarrow \infty$ ($t \rightarrow \infty$), thus $\hat{\boldsymbol{\mu}}(t) \rightarrow \boldsymbol{\mu}$ ($t \rightarrow \infty$) by law of large numbers.
727 Therefore, $\mathbf{w}^{s_0}(\hat{\boldsymbol{\mu}}(t)) \rightarrow \mathbf{w}^{s_0}(\boldsymbol{\mu})$ ($t \rightarrow \infty$) holds by the continuity of the function $\mathbf{w}^{s_0}(\cdot)$ in the
728 neighborhood of $\boldsymbol{\mu}$. Then, there exists a positive integer t_0 such that

$$|w_{\ell}^{s_0}(\hat{\boldsymbol{\mu}}(t)) - w_{\ell}^{s_0}(\boldsymbol{\mu})| \leq \epsilon \text{ for all } t \geq t_0 \text{ and } \ell \in \mathcal{L}(\mathcal{T}).$$

729 Thus, by Lemma C.17,

$$\frac{N_{\ell}(t)}{t} \rightarrow w_{\ell}^{s_0}(\boldsymbol{\mu}) \text{ } (t \rightarrow \infty) \text{ for all } \ell \in \mathcal{L}(\mathcal{T}).$$

730 □

731 **D Proof for Sample Complexity of Algorithm 1 (Proof of Theorem 5.4)**

732 The proof is conducted along the same lines as the proof of Theorem 14 in Garivier and Kaufmann
733 [2016].

734 For $\epsilon > 0$, define

$$\mathcal{I}_\epsilon = [\mu_\ell - \xi(\epsilon), \mu_\ell + \xi(\epsilon)]^{|\mathcal{L}(\mathcal{T})|}$$

735 using $\xi(\epsilon) > 0$ that satisfies

$$\forall \boldsymbol{\lambda} \in \mathcal{I}_\epsilon, \max_{\ell \in \mathcal{L}(\mathcal{T})} |w_\ell^{s_0}(\boldsymbol{\lambda}) - w_\ell^{s_0}(\boldsymbol{\mu})| < \epsilon.$$

736 Such $\xi(\epsilon)$ exists by Lemma C.9.

737 Define Event $\mathcal{E}_T(\epsilon)$ as

$$\mathcal{E}_T(\epsilon) = \bigcap_{t=T^{1/4}}^T (\hat{\boldsymbol{\mu}}(t) \in \mathcal{I}_\epsilon)$$

738 for any $T \in \mathbb{N}$.

739 The following Corollary D.1 and Lemma D.2 are used to prove the theorem.

740 Corollary D.1 is the RD-Tracking-TMCTS version of Lemma 19 [Garivier and Kaufmann, 2016],
741 which holds for the forced exploration of D-Tracking.

742 **Corollary D.1** (Corollary of Lemma 19 [Garivier and Kaufmann, 2016]). *Let $\mathcal{E}_T^c(\epsilon)$ denote the*
743 *complementary event of event $\mathcal{E}_T(\epsilon)$. Then, inequality*

$$\mathbb{P}(\mathcal{E}_T^c(\epsilon)) < BT \exp(-CT^{1/8})$$

744 *holds for any sampling rule that ensures $N_\ell(t) \geq \sqrt{t} - |\mathcal{L}(\mathcal{T})|$ for $t \geq T^{1/4}$, where*

$$B = \sum_{\ell \in \mathcal{L}(\mathcal{T})} \left(\frac{e^{|\mathcal{L}(\mathcal{T})|d(\mu_\ell - \xi(\epsilon), \mu_\ell)}}{1 - e^{|\mathcal{L}(\mathcal{T})|d(\mu_\ell - \xi(\epsilon), \mu_\ell)}} + \frac{e^{|\mathcal{L}(\mathcal{T})|d(\mu_\ell + \xi(\epsilon), \mu_\ell)}}{1 - e^{|\mathcal{L}(\mathcal{T})|d(\mu_\ell + \xi(\epsilon), \mu_\ell)}} \right) \text{ and}$$

$$C = \min_{\ell \in \mathcal{L}(\mathcal{T})} \min\{d(\mu_\ell - \xi(\epsilon), \mu_\ell), d(\mu_\ell + \xi(\epsilon), \mu_\ell)\}.$$

745 The following lemma can be obtained by the definition of $\mathcal{I}_\epsilon$ and Lemma C.17 (and Remark C.18).

746 **Lemma D.2** (RD-Tracking-TMCTS version of Lemma 20 of [Garivier and Kaufmann, 2016]). *For*
747 *RD-Tracking-TMCTS, there exists T_ϵ such that, for any $T \geq T_\epsilon$, $\mathcal{E}_T(\epsilon)$ implies the following.*

$$\forall t \in [\sqrt{T}, T] \cap \mathbb{N}, \forall \ell \in \mathcal{L}(\mathcal{T}), \left| \frac{N_\ell(t)}{t} - w_\ell^{s_0}(\boldsymbol{\mu}) \right| \leq 2(|\mathcal{L}(\mathcal{T})| - 1)\epsilon.$$

748 *Proof of Theorem 5.4.* Let

$$g(\boldsymbol{\mu}', \boldsymbol{w}) = \inf_{\boldsymbol{\lambda} \in \text{Alt}(\boldsymbol{\mu}')} \sum_{\ell \in \mathcal{L}(\mathcal{T})} w_\ell d(\mu'_\ell, \lambda_\ell).$$

749 Then, using function g , the GLR statistic $Z_{s_0}(t)$ can be represented as

$$\begin{aligned} Z_{s_0}(t) &= \inf_{\boldsymbol{\lambda} \in \text{Alt}(\hat{\boldsymbol{\mu}}(t))} \sum_{\ell \in \mathcal{L}(\mathcal{T})} N_\ell(t) d(\hat{\mu}_\ell(t), \lambda_\ell) \\ &= t \inf_{\boldsymbol{\lambda} \in \text{Alt}(\hat{\boldsymbol{\mu}}(t))} \sum_{\ell \in \mathcal{L}(\mathcal{T})} \frac{N_\ell(t)}{t} d(\hat{\mu}_\ell(t), \lambda_\ell) \\ &= t g\left(\hat{\boldsymbol{\mu}}(t), \left(\frac{N_\ell(t)}{t}\right)_{\ell \in \mathcal{L}(\mathcal{T})}\right). \end{aligned}$$

750 Define $d_{s_0}^\epsilon(\boldsymbol{\mu})$ as

$$d_{s_0}^\epsilon(\boldsymbol{\mu}) := \inf_{\substack{\boldsymbol{\mu}' \in \mathcal{I}_\epsilon \\ \boldsymbol{w}' : \|\boldsymbol{w}' - \boldsymbol{w}^{s_0}(\boldsymbol{\mu})\| < 2(|\mathcal{L}(\mathcal{T})| - 1)\epsilon}} g(\boldsymbol{\mu}', \boldsymbol{w}')$$

751 Then, there exists T_ϵ such that, on the event $\mathcal{E}_T(\epsilon)$ and for $T \geq T_\epsilon$,

$$Z_{s_0}(t) \geq td_{s_0}^\epsilon(\boldsymbol{\mu}).$$

752 holds for every $\sqrt{T} \leq t \leq T$ by Lemma D.2.

753 Thus, for $T \geq T_\epsilon$ and on $\mathcal{E}_T(\epsilon)$,

$$\begin{aligned} \min(\tau_\delta, T) &\leq \sqrt{T} + \sum_{t=\sqrt{T}}^T \mathbb{1}\{\tau_\delta > t\} \\ &\leq \sqrt{T} + \sum_{t=\sqrt{T}}^T \mathbb{1}\{Z_{s_0}(t) < \beta(t, \delta)\} \\ &\leq \sqrt{T} + \sum_{t=\sqrt{T}}^T \mathbb{1}\{td_{s_0}^\epsilon(\boldsymbol{\mu}) < \beta(T, \delta)\} \\ &\leq \sqrt{T} + \frac{\beta(T, \delta)}{d_{s_0}^\epsilon(\boldsymbol{\mu})} \end{aligned}$$

754 holds. Let $T_0(\delta)$ be

$$T_0(\delta) := \inf \left\{ T \in \mathbb{N} : \sqrt{T} + \frac{\beta(T, \delta)}{d_{s_0}^\epsilon(\boldsymbol{\mu})} \leq T \right\}.$$

755 Then, for every $T \geq \max(T_0(\delta), T_\epsilon)$, we have $\mathcal{E}_T(\epsilon) \subseteq (\tau_\delta \leq T)$, which implies

$$\mathbb{P}_\mu(\tau_\delta > T) \leq \mathbb{P}(\mathcal{E}_T^c) \leq BT \exp(-CT^{1/8})$$

756 using Corollary D.1. Therefore,

$$\begin{aligned} \mathbb{E}[\tau_\delta] &= \sum_{T=0}^{\infty} \mathbb{P}_\mu(\tau_\delta > T) \\ &\leq \max(T_0(\delta), T_\epsilon) + \sum_{T=\max(T_0(\delta), T_\epsilon)}^{\infty} \mathbb{P}_\mu(\tau_\delta > T) \\ &\leq T_0(\delta) + T_\epsilon + \sum_{T=1}^{\infty} BT \exp(-CT^{1/8}). \end{aligned}$$

757 Note that the sum of the last two terms, $T_\epsilon + \sum_{T=1}^{\infty} BT \exp(-CT^{1/8})$, is a constant depending on ϵ .

758 We now analyze an upper bound on $T_0(\delta)$. Let $\eta > 0$ and define constant $C(\eta)$ as

$$C(\eta) = \inf \left\{ T \in \mathbb{N} : T - \sqrt{T} \geq T/(1 + \eta) \right\}.$$

759 Then,

$$\begin{aligned} T_0(\delta) &\leq C(\eta) + \inf \left\{ T \in \mathbb{N} : \frac{\beta(T, \delta)}{d_{s_0}^\epsilon(\boldsymbol{\mu})} \leq \frac{T}{(1 + \eta)} \right\} \\ &= C(\eta) + \frac{(1 + \eta) \ln(1/\delta)}{d_{s_0}^\epsilon(\boldsymbol{\mu})} + o(\ln(1/\delta)) \end{aligned} \quad (\text{by Lemma C.8})$$

760 holds. Thus, we have

$$\mathbb{E}[\tau_\delta] \leq \frac{(1 + \eta) \ln(1/\delta)}{d_{s_0}^\epsilon(\boldsymbol{\mu})} + o(\ln(1/\delta)),$$

761 which yields, for every $\eta > 0$ and $\epsilon > 0$,

$$\limsup_{\delta \rightarrow +0} \frac{\mathbb{E}_\mu[\tau_\delta]}{\log(1/\delta)} \leq \frac{1 + \eta}{d_{s_0}^\epsilon(\boldsymbol{\mu})}.$$

762 Since

$$\lim_{\epsilon \rightarrow +0} d_{s_0}^\epsilon(\mu) = d_{s_0}(\mu)$$

763 holds by continuity of g and definition of w^{s_0} , we have

$$\limsup_{\delta \rightarrow +0} \frac{\mathbb{E}_\mu[\tau_\delta]}{\log(1/\delta)} \leq \frac{1}{d_{s_0}(\mu)}$$

764 by letting η and ϵ go to zero.

765

□

766 E Time Complexity Analysis

767 E.1 Heap Data Structure

768 Let $\{(x_i, i)\}_{i=1, \dots, N}$ be a set of N couples, with $x_i \in \mathbb{R}$. We consider a heap data structure, IHeap,
769 that supports the following operations on this set.

770 **heapify**($\{(x_i, i)\}_{i=1, \dots, N}$): Initialize the heap data structure with the given set $\{(i, x_i)\}_{i=1, \dots, N}$ in
771 $O(N)$ time.

772 **peek**: Returns (x_{i^*}, i^*) with $i^* = \arg \min_i x_i$ in $O(1)$ time.

773 **rewrite**(v, i): Replace the value associated with index i , namely (x_i, i) , with (v, i) and maintain the
774 heap in $O(\log N)$ time.

775 This data structure can be implemented with a heap ordered by the values x_i . By maintaining an
776 auxiliary array $\text{pos}[i]$ storing the heap position of each element, a $\text{rewrite}(v, i)$ operation can be
777 performed in $O(\log N)$ time via a constant-time update followed by at most $\log N$ heap adjustments.

778 Similarly, we consider another heap data structure DHeap. It supports the same operations as IHeap,
779 except that the peek operation returns the arg max element instead of the arg min.

780 For any couple $A = (a, b)$, $A[0]$ and $A[1]$ denote a and b , respectively. For example, we use
781 $\text{IHeap-}\tilde{d}_s$ as IHeap for $\tilde{d}_c(\hat{\mu}(t))$ ($c \in \mathcal{C}(s)$), and $\text{IHeap-}\tilde{d}_s.\text{peek}[0]$ and $\text{IHeap-}\tilde{d}_s.\text{peek}[1]$ denote
782 $\min_{c \in \mathcal{C}} \tilde{d}_c(\hat{\mu}(t))$ and $\arg \min_{c \in \mathcal{C}} \tilde{d}_c(\hat{\mu}(t))$, respectively.

783 E.2 Efficient implementation of RD-Tracking-TMCTS sampling rule

784 First, we consider the forced exploration step (the first case of Eq. (11)). Maintaining an IHeap with
785 the set $\{(N_\ell(t), \ell)\}_{\ell \in \mathcal{L}(\mathcal{T})}$, we can obtain $\arg \min_{\ell \in \mathcal{L}(\mathcal{T})} N_\ell(t)$.

786 Next, we consider the ratio-based tracking step (the second case of Eq. (11)). To compute $I(t)$, we
787 introduce a subproblem that computes

$$\text{RD}_s(t) = \left(\frac{1}{\tilde{d}_{I_s(t)}(\hat{\mu}(t)) N_{I_s(t)}(t)}, I_s(t) \right),$$

788 for each node $s \in \mathcal{L}(\mathcal{T})$, where

$$I_s(t) = \arg \max_{\ell \in \mathcal{D}(s)} w_\ell^s(\hat{\mu}(t)) / N_\ell(t).$$

789 Note that $I_{s_0}(t) = I(t)$. Then, the following theorem holds.

790 **Theorem E.1.** For each $s \in S$, $\text{RD}_s(t)$ can be calculated by Recursive Formula 2.

791 *Proof.* We prove the theorem by mathematical induction on the height of the tree node s .

792 Consider a height-0 node s . In this case, node s is a leaf node and $\mathcal{D}(s) = \{s\}$. Thus,

$$\text{RD}_s(t) = \begin{cases} \left(\frac{1}{\tilde{d}_s(\hat{\mu}(t)) N_s(t)}, s \right) & (\tilde{d}_s(\hat{\mu}(t)) \neq 0) \\ (+\infty, s) & (\tilde{d}_s(\hat{\mu}(t)) = 0) \end{cases}$$

$$\text{RD}_s(t) = \begin{cases} \left(\frac{1}{\tilde{d}_s(\hat{\mu}(t))N_s(t)}, s \right) & (s \in \mathcal{L}(T), \tilde{d}_s(\hat{\mu}(t)) \neq 0) \\ (+\infty, s) & (s \in \mathcal{L}(T), \tilde{d}_s(\hat{\mu}(t)) = 0) \\ \text{RD}_{c^+(s)}(t) & \left(\begin{array}{l} L(s) = \text{'MAX'}, \\ \tilde{d}_c(\mu) \geq 0 \text{ for some } c \in \mathcal{C}(s) \end{array} \right) \\ \arg \min_{A \in \{\text{RD}_c(t) | c \in \mathcal{C}(s)\}} A[0] & \left(\begin{array}{l} L(s) = \text{'MAX'}, \\ \tilde{d}_c(\mu) < 0 \text{ for all } c \in \mathcal{C}(s) \end{array} \right) \\ \arg \max_{A \in \{\text{RD}_c(t) | c \in \mathcal{C}(s)\}} A[0] & \left(\begin{array}{l} L(s) = \text{'MIN'}, \\ \tilde{d}_c(\mu) > 0 \text{ for all } c \in \mathcal{C}(s) \end{array} \right) \\ \text{RD}_{c^-(s)}(t) & \left(\begin{array}{l} L(s) = \text{'MIN'}, \\ \tilde{d}_c(\mu) \leq 0 \text{ for some } c \in \mathcal{C}(s) \end{array} \right) \end{cases}$$

793 holds, which coincides with $\text{RD}_s(t)$ calculated by Recursive Formula 2.

794 Assume that $\text{RD}_s(t)$ can be calculated by Recursive Formula 2 for a height- k node s . Consider a
795 height- $k + 1$ node s .

796 **The case with $L(s) = \text{'MAX'}$ and $a_s(\hat{\mu}(t)) = \text{'win'}$:**

797 Since $L(s) = \text{'MAX'}$ and $a_s(\hat{\mu}(t)) = \text{'win'}$, there exists $c \in \mathcal{C}(s)$ that satisfies $a_c(\hat{\mu}(t)) = \text{'win'}$
798 and $\tilde{d}_c(\hat{\mu}(t)) \geq 0$. From Recursive Formula 1 we have,

$$I_s(t) = \arg \max_{\ell \in \mathcal{D}(s)} w_\ell^s(\hat{\mu}(t))/N_\ell(t) = \arg \max_{\ell \in \mathcal{D}(c^+(s))} w_\ell^{c^+(s)}(\hat{\mu}(t))/N_\ell(t) = I_{c^+(s)}(t)$$

799 where $c^+(s) = \arg \max_{c \in \mathcal{C}(s)} \tilde{d}_c(\hat{\mu}(t))$. Therefore, $\text{RD}_s(t) = \text{RD}_{c^+(s)}(t)$. Thus, $\text{RD}_s(t)$ can be calcu-
800 lated by Recursive Formula 2.

801 **The case with $L(s) = \text{'MIN'}$ and $a_s(\hat{\mu}(t)) = \text{'lose'}$:**

802 By the same argument,

$$I_s(t) = \arg \max_{\ell \in \mathcal{D}(s)} w_\ell^s(\hat{\mu}(t))/N_\ell(t) = \arg \max_{\ell \in \mathcal{D}(c^-(s))} w_\ell^{c^-(s)}(\hat{\mu}(t))/N_\ell(t) = I_{c^-(s)}(t)$$

803 where $c^-(s) = \arg \min_{c \in \mathcal{C}(s)} \tilde{d}_c(\hat{\mu}(t))$. Therefore, $\text{RD}_s(t) = \text{RD}_{c^-(s)}(t)$. Thus, $\text{RD}_s(t)$ can be calcu-
804 lated by Recursive Formula 2.

805 **The case with $L(s) = \text{'MAX'}$ and $a_s(\hat{\mu}(t)) = \text{'lose'}$:**

806 Since $L(s) = \text{'MAX'}$ and $a_s(\hat{\mu}(t)) = \text{'lose'}$, all $c \in \mathcal{C}(s)$ satisfies $a_c(\hat{\mu}(t)) = \text{'lose'}$ and

807 $\tilde{d}_c(\hat{\mu}(t)) < 0$. Since $w_\ell^s(\hat{\mu}(t)) = \frac{d_s(\hat{\mu}(t))}{d(\hat{\mu}_\ell(t), \theta)}$ holds for $w_\ell^s(\hat{\mu}(t)) > 0$ by Proposition C.4,

$$\begin{aligned}
\max_{\ell \in \mathcal{D}(s)} w_\ell^s(\hat{\mu}(t))/N_\ell(t) &= \max_{c \in \mathcal{C}(s)} \max_{\ell \in \mathcal{D}(c)} w_\ell^s(\hat{\mu}(t))/N_\ell(t) \\
&= \max_{c \in \mathcal{C}(s)} \max_{\ell \in \mathcal{D}(c), w_\ell^s(\hat{\mu}(t)) > 0} \frac{d_s(\hat{\mu}(t))}{d(\hat{\mu}_\ell(t), \theta)N_\ell(t)} \\
&= d_s(\hat{\mu}(t)) \max_{c \in \mathcal{C}(s)} \max_{\ell \in \mathcal{D}(c), w_\ell^s(\hat{\mu}(t)) > 0} \frac{1}{d(\hat{\mu}_\ell(t), \theta)N_\ell(t)} \\
&= d_s(\hat{\mu}(t)) \min_{c \in \mathcal{C}(s)} \min_{\ell \in \mathcal{D}(c), w_\ell^s(\hat{\mu}(t)) > 0} \frac{1}{(-d(\hat{\mu}_\ell(t), \theta))N_\ell(t)} \\
&= d_s(\hat{\mu}(t)) \min_{c \in \mathcal{C}(s)} \min_{\ell \in \mathcal{D}(c), w_\ell^s(\hat{\mu}(t)) > 0} \frac{1}{\tilde{d}(\hat{\mu}_\ell(t))N_\ell(t)} \\
&= d_s(\hat{\mu}(t)) \min_{c \in \mathcal{C}(s)} \frac{1}{\tilde{d}(\hat{\mu}_{I_c(t)}(t))N_{I_c(t)}(t)} \\
&= d_s(\hat{\mu}(t)) \min_{A \in \{\text{RD}_c(t) | c \in \mathcal{C}(s)\}} A[0]
\end{aligned}$$

808 holds, which means $\text{RD}_s(t) = \arg \min_{A \in \{\text{RD}_c(t) | c \in \mathcal{C}(s)\}} A[0]$. Thus, $\text{RD}_s(t)$ can be calculated by Recursive

809 Formula 2.

810 **The case with $L(s) = \text{'MIN'}$ and $a_s(\hat{\mu}(t)) = \text{'win'}$:**

811 Similarly,

$$\begin{aligned}
\max_{\ell \in \mathcal{D}(s)} w_\ell^s(\hat{\mu}(t))/N_\ell(t) &= \max_{c \in \mathcal{C}(s)} \max_{\ell \in \mathcal{D}(c)} w_\ell^s(\hat{\mu}(t))/N_\ell(t) \\
&= d_s(\hat{\mu}(t)) \max_{A \in \{\text{RD}_c(t) | c \in \mathcal{C}(s)\}} A[0]
\end{aligned}$$

812 holds, which means $\text{RD}_s(t) = \arg \max_{A \in \{\text{RD}_c(t) | c \in \mathcal{C}(s)\}} A[0]$. Thus, $\text{RD}_s(t)$ can be calculated by Recursive

813 Formula 2. □

814 E.3 Algorithm detail

Algorithm 2 The detailed pseudocode of RD-Tracking-TMCTS

Require: $\mathcal{T}$: tree, θ : threshold, δ : confidence parameter

Ensure: Return $a_{s_0}(\mu) \in \{\text{'win'}, \text{'lose'}\}$ correctly with prob. at least $1 - \delta$

```

1:  $t \leftarrow 1$ 
2: for each  $\ell \in \mathcal{L}(\mathcal{T})$  do
3:   Draw  $\ell$  and observe reward  $r(t)$ .
4:    $N_\ell \leftarrow 1, \hat{\mu}_\ell \leftarrow r(t), t \leftarrow t + 1$ 
5:  $(\tilde{d}_{s_0}, \tilde{Z}_{s_0}, \text{RD}_{s_0}) \leftarrow \text{initialization}(s_0)$  (Algorithm 3)
6:  $\text{IHeap-}N.\text{heapify}(\{(N_\ell, \ell)\})$ 
7: loop
8:   if  $\text{IHeap-}N.\text{peek}[0] < \sqrt{t} - |\mathcal{L}(\mathcal{T})|/2$  then
9:      $I(t) \leftarrow \text{IHeap-}N.\text{peek}[1]$ 
10:  else
11:     $I(t) \leftarrow \text{RD}_{s_0}[1]$ 
12:  Observe reward  $r(t)$  from leaf  $I(t)$ 
13:   $\text{update\_estimation}(I(t), r(t))$  (Algorithm 4)
14:  if  $|\tilde{Z}_{s_0}(t)| \geq \beta(t, \delta)$  then
15:    return  $\begin{cases} \text{'win'} & (\tilde{Z}_{s_0}(t) \geq 0) \\ \text{'lose'} & (\tilde{Z}_{s_0}(t) < 0) \end{cases}$ 
16:   $t \leftarrow t + 1$ 

```

815 The detailed pseudocode of RD-Tracking-TMCTS (Algorithm 1) is shown in Algorithm 2.

Algorithm 3 initialization

Require: s : node

```

1: if  $s \in \mathcal{L}(\mathcal{T})$  then
2:    $\tilde{d}_s \leftarrow \begin{cases} d(\hat{\mu}_s(t), \theta) & (\hat{\mu}_s(t) \geq \theta) \\ -d(\hat{\mu}_s(t), \theta) & (\hat{\mu}_s(t) < \theta) \end{cases}$ 
3:    $\tilde{Z}_s \leftarrow N_s(t)\tilde{d}_s$ 
4:    $\text{RD}_s \leftarrow (1/\tilde{d}_s N_s, s)$ 
5:   return  $(\tilde{d}_s, \tilde{Z}_s, \text{RD}_s)$ 
6: else
7:    $i \leftarrow 0, A_{\tilde{d}}, A_{\tilde{Z}}, A_{\text{RD}} \leftarrow []$ 
8:   for  $c \in \mathcal{C}(s)$  do
9:      $(\tilde{d}_c, \tilde{Z}_c, \text{RD}_c) \leftarrow \text{initialization}(c)$ 
10:     $A_{\tilde{d}}[i] \leftarrow (\tilde{d}_c, c)$ 
11:     $A_{\tilde{Z}}[i] \leftarrow (\tilde{Z}_c, c)$ 
12:     $A_{\text{RD}}[i] \leftarrow \text{RD}_c$ 
13:     $i \leftarrow i + 1$ 
14:   if  $L(s) = \text{'MAX'}$  then
15:      $\text{RSR}_s \leftarrow 1 / \sum_{c \in \mathcal{C}(s), \tilde{d}_c < 0} (1/\tilde{d}_c)$ 
16:      $\text{SZ}_s \leftarrow \sum_{c \in \mathcal{C}(s), \tilde{Z}_c \geq 0} \tilde{Z}_c$ 
17:      $\text{DHeap-}\tilde{d}_s.\text{heapify}(A_{\tilde{d}})$ 
18:      $\text{DHeap-}\tilde{Z}_s.\text{heapify}(A_{\tilde{Z}})$ 
19:      $\text{IHeap-RD}_s.\text{heapify}(A_{\text{RD}})$ 
20:     if  $\text{DHeap-}\tilde{d}_s.\text{peek}[0] \geq 0$  then
21:        $\tilde{d}_s \leftarrow \text{DHeap-}\tilde{d}_s.\text{peek}[0]$ 
22:        $\tilde{Z}_s \leftarrow \text{SZ}_s$ 
23:        $\text{RD}_s \leftarrow \text{RD}_{\text{DHeap-}\tilde{d}_s.\text{peek}[1]}$ 
24:     else
25:        $\tilde{d}_s \leftarrow \text{RSR}_s$ 
26:        $\tilde{Z}_s \leftarrow \text{DHeap-}\tilde{Z}_s.\text{peek}[0]$ 
27:        $\text{RD}_s \leftarrow \text{IHeap-RD}_s.\text{peek}$ 
28:   else
29:      $\text{RSR}_s \leftarrow 1 / \sum_{c \in \mathcal{C}(s), \tilde{d}_c > 0} (1/\tilde{d}_c)$ 
30:      $\text{SZ}_s \leftarrow \sum_{c \in \mathcal{C}(s), \tilde{Z}_c < 0} \tilde{Z}_c$ 
31:      $\text{IHeap-}\tilde{d}_s.\text{heapify}(A_{\tilde{d}})$ 
32:      $\text{IHeap-}\tilde{Z}_s.\text{heapify}(A_{\tilde{Z}})$ 
33:      $\text{DHeap-RD}_s.\text{heapify}(A_{\text{RD}})$ 
34:     if  $\text{IHeap-}\tilde{d}_s.\text{peek}[0] < 0$  then
35:        $\tilde{d}_s \leftarrow \text{IHeap-}\tilde{d}_s.\text{peek}[0]$ 
36:        $\tilde{Z}_s \leftarrow \text{SZ}_s$ 
37:        $\text{RD}_s \leftarrow \text{RD}_{\text{IHeap-}\tilde{d}_s.\text{peek}[1]}$ 
38:     else
39:        $\tilde{d}_s \leftarrow \text{RSR}_s$ 
40:        $\tilde{Z}_s \leftarrow \text{IHeap-}\tilde{Z}_s.\text{peek}[0]$ 
41:        $\text{RD}_s \leftarrow \text{DHeap-RD}_s.\text{peek}$ 
42:   return  $(\tilde{d}_s, \tilde{Z}_s, \text{RD}_s)$ 

```

816 At the first $\mathcal{L}(\mathcal{T})$ rounds, each leaf $\ell \in \mathcal{L}(\mathcal{T})$ is drawn once, and N_ℓ and $\hat{\mu}_\ell$ are set to 1 and the
 817 obtained reward $r(t)$, respectively (Lines 2-4). Then, for each internal node $s \in S$, $(\tilde{d}_s, \tilde{Z}_s, \text{RD}_s)$
 818 is calculated and the heaps on $\{(\tilde{d}_c, c) \mid c \in \mathcal{C}(s)\}$, $\{(\tilde{Z}_c, c) \mid c \in \mathcal{C}(s)\}$ and $\{\text{RD}_c \mid c \in \mathcal{C}(s)\}$ is
 819 constructed by calling $\text{initialization}(s_0)$ at Line 5. IHeap on $\{(N_\ell, \ell) \mid \ell \in \mathcal{L}(\mathcal{T})\}$, IHeap- N , is also
 820 constructed at Line 6. At round $t = \mathcal{L}(\mathcal{T}) + 1, \mathcal{L}(\mathcal{T}) + 2, \dots$, the codes inside the loop (Lines 8-16)

Algorithm 4 update_estimation

Require: ℓ : leaf, r : reward
Ensure: update $\hat{\mu}_\ell$, N_ℓ , heaps correctly

```
1:  $\text{prev-}\tilde{d}_c \leftarrow \tilde{d}_\ell$ ,  $\text{prev-}\tilde{Z}_c \leftarrow \tilde{Z}_\ell$ 
2:  $\hat{\mu}_\ell \leftarrow \frac{\hat{\mu}_\ell N_\ell + r}{N_\ell + 1}$ ,  $N_\ell \leftarrow N_\ell + 1$ 
3:  $\tilde{d}_\ell \leftarrow \begin{cases} d(\hat{\mu}_\ell, \theta) & (\hat{\mu}_\ell \geq \theta) \\ -d(\hat{\mu}_\ell, \theta) & (\hat{\mu}_\ell < \theta) \end{cases}$ 
4:  $\tilde{Z}_\ell \leftarrow N_\ell \tilde{d}_\ell$ ,  $\text{RD}_\ell \leftarrow (1/\tilde{Z}_\ell, \ell)$ 
5:  $c \leftarrow \ell$ ,  $s \leftarrow \text{parent node of } c$ 
6: repeat
7:   if  $L(s) = \text{'MAX'}$  then
8:     if  $\text{prev-}\tilde{d}_c < 0$  then
9:        $\text{RSR}_s \leftarrow 1/(1/\text{RSR}_s - 1/\text{prev-}\tilde{d}_c)$ 
10:    else
11:       $\text{SZ}_s \leftarrow \text{SZ}_s - \text{prev-}\tilde{Z}_c$ 
12:    if  $\tilde{d}_c < 0$  then
13:       $\text{RSR}_s \leftarrow 1/(1/\text{RSR}_s + 1/\tilde{d}_c)$ 
14:    else
15:       $\text{SZ}_s \leftarrow \text{SZ}_s + \tilde{Z}_c$ 
16:     $\text{prev-}\tilde{d}_c \leftarrow \tilde{d}_s$ ,  $\text{prev-}\tilde{Z}_c \leftarrow \tilde{Z}_s$ 
17:     $\text{DHeap-}\tilde{d}_s.\text{rewrite}(\tilde{d}_c, c)$ 
18:     $\text{DHeap-}\tilde{Z}_s.\text{rewrite}(\tilde{Z}_c, c)$ 
19:     $\text{IHeap-RD}_s.\text{rewrite}(\text{RD}_c, c)$ 
20:    if  $\text{DHeap-}\tilde{d}_s.\text{peek}[0] \geq 0$  then
21:       $\tilde{d}_s \leftarrow \text{DHeap-}\tilde{d}_s.\text{peek}[0]$ 
22:       $\tilde{Z}_s \leftarrow \text{SZ}_s$ 
23:       $\text{RD}_s \leftarrow \text{RD}_{\text{DHeap-}\tilde{d}_s.\text{peek}[1]}$ 
24:    else
25:       $\tilde{d}_s \leftarrow \text{RSR}_s$ 
26:       $\tilde{Z}_s \leftarrow \text{DHeap-}\tilde{Z}_s.\text{peek}[0]$ 
27:       $\text{RD}_s \leftarrow \text{IHeap-RD}_s.\text{peek}$ 
28:  else
29:    if  $\text{prev-}\tilde{d}_c > 0$  then
30:       $\text{RSR}_s \leftarrow 1/(1/\text{RSR}_s - 1/\text{prev-}\tilde{d}_c)$ 
31:    else
32:       $\text{SZ}_s \leftarrow \text{SZ}_s - \text{prev-}\tilde{Z}_c$ 
33:    if  $\tilde{d}_c > 0$  then
34:       $\text{RSR}_s \leftarrow 1/(1/\text{RSR}_s + 1/\tilde{d}_c)$ 
35:    else
36:       $\text{SZ}_s \leftarrow \text{SZ}_s + \tilde{Z}_s$ 
37:     $\text{prev-}\tilde{d}_c \leftarrow \tilde{d}_s$ ,  $\text{prev-}\tilde{Z}_c \leftarrow \tilde{Z}_s$ 
38:     $\text{IHeap-}\tilde{d}_s.\text{rewrite}(\tilde{d}_c, c)$ 
39:     $\text{IHeap-}\tilde{Z}_s.\text{rewrite}(\tilde{Z}_c, c)$ 
40:     $\text{DHeap-RD}_s.\text{rewrite}(\text{RD}_c, c)$ 
41:    if  $\text{IHeap-}\tilde{d}_s.\text{peek}[0] < 0$  then
42:       $\tilde{d}_s \leftarrow \text{IHeap-}\tilde{d}_s.\text{peek}[0]$ 
43:       $\tilde{Z}_s \leftarrow \text{SZ}_s$ 
44:       $\text{RD}_s \leftarrow \text{RD}_{\text{IHeap-}\tilde{d}_s.\text{peek}[1]}$ 
45:    else
46:       $\tilde{d}_s \leftarrow \text{RSR}_s$ 
47:       $\tilde{Z}_s \leftarrow \text{IHeap-}\tilde{Z}_s.\text{peek}[0]$ 
48:       $\text{RD}_s \leftarrow \text{DHeap-RD}_s.\text{peek}$ 
49:     $c \leftarrow s$ ,  $s \leftarrow \text{parent node of } c$ 
50: until  $s = \text{NULL}$ 
```

821 is executed. In the loop, leaf $I(t)$ is selected by the sampling rule (11) at Lines 8-11, which is done
822 only by referring $\text{IHeap-}N.\text{peek}[1]$ in the forced exploration case and $\text{RD}_{s_0}[1]$ otherwise. After
823 observing the reward $r(t)$ from leaf $I(t)$ (Line 12), $(\tilde{d}_s, \tilde{Z}_s, \text{RD}_s)$ is updated only for each node
824 $s = s_0, s_1, \dots, I(t)$ on the path from s_0 to $I(t)$ and the rewrite operation is executed for the heaps of
825 such s by calling $\text{update_estimation}(I(t), r(t))$ (Line 13). Using the updated $\tilde{Z}_{s_0}(t)$, the stopping
826 condition is checked at Line 14, and return 'win' or 'lose' (Line 15) if the condition is satisfied.

Algorithm ‘initialization’ is shown in Algorithm 3. In this algorithm, $(\tilde{d}_s, \tilde{Z}_s, \text{RD}_s)$ is recursively calculated for each $s \in S$. For s with $L(s) = \text{‘MAX’}$, DHeaps are constructed on $\{(\tilde{d}_c, c) \mid c \in \mathcal{C}(s)\}$ and $\{(\tilde{Z}_c, c) \mid c \in \mathcal{C}(s)\}$ and an IHeap is constructed on $\{(\text{RD}_c, c) \mid c \in \mathcal{C}(s)\}$ by executing heapify operations. For s with $L(s) = \text{‘MIN’}$, IHeaps are constructed on $\{(\tilde{d}_c, c) \mid c \in \mathcal{C}(s)\}$ and $\{(\tilde{Z}_c, c) \mid c \in \mathcal{C}(s)\}$ and a DHeap is constructed on $\{(\text{RD}_c, c) \mid c \in \mathcal{C}(s)\}$ by executing heapify operations. Note that SZ_s (“sum of Z ”) and RSR_s (“reciprocal sum of the reciprocal”) must be maintained to calculate $\tilde{Z}_s$ and $\tilde{d}_s$, respectively, in $O(1)$ time.

Algorithm ‘update_estimation’ is shown in Algorithm 4. This algorithm first updates statistics $(\hat{\mu}_\ell, N_\ell, \tilde{d}_\ell, \tilde{Z}_\ell, \text{RD}_\ell)$ of the leaf ℓ , and updates $(\tilde{d}_s, \tilde{Z}_s, \text{RD}_s)$ of each node s on the path from s_0 to ℓ by repeatedly going up the path. The updated values are reflected to the maintained heaps for nodes s by executing rewrite operations. RSR_s and SZ_s for such s are also maintained by subtracting the previous value and adding the updated value.

Proof of Theorem 5.5. Our tracking algorithm maintains the statistics of $|\text{SI}|$ nodes, and its space complexity is $O(|\text{SI}|)$.

Consider the first $\mathcal{L}(\mathcal{T})$ rounds. The first for-loop in Algorithm 2 takes $O(\mathcal{L}(\mathcal{T}))$ time. Algorithm initialization(s_0) executes heapify and peek operations for the $|\mathcal{C}(s)|$ -sized array for each $s \in S$, thus it takes $O(S) = O(\mathcal{L}(\mathcal{T}))$ time. Therefore, the amortized time per round during the first $\mathcal{L}(\mathcal{T})$ rounds is $O(1)$.

The codes inside the loop is executed in any round after the first $\mathcal{L}(\mathcal{T})$ rounds, and $O(1)$ time is needed except execution time of Algorithm ‘update_estimation’. Since it executes rewrite and peek operations for nodes s on the path from s_0 to leaf ℓ , it takes $O(\mathcal{D}(\mathcal{T}) \log \mathcal{W}(\mathcal{T}))$ time. Thus, totally $O(\mathcal{D}(\mathcal{T}) \log \mathcal{W}(\mathcal{T}))$ time per round is enough for the codes inside the loop.

Therefore, RD-Tracking-TMCTS runs in $O(\mathcal{D}(\mathcal{T}) \log \mathcal{W}(\mathcal{T}))$ amortized time per round. $\square$

F Non-unique Optimal Proportions

When the optimal proportions oscillate, the algorithm may fail to converge its sampling proportions to the optimal proportions [Degenne and Koolen, 2019]. However, in some cases, we can show that our method works in non-unique optimal proportions.

Example F.1. We consider $a_{s_0}(\mu) = \text{‘win’}$ and $L(s_0) = \text{‘MAX’}$. If two (or more) depth-1 nodes s have an identical value $d_s(\mu) = \max_{s' \in \mathcal{C}(s_0)} d_{s'}(\mu)$, $w^{s_0}(\hat{\mu}(t))$ oscillates between $w^s(\mu)$. In this setting, the $w^s(\mu)$ are mutually exclusive, and $\{\ell : w^s(\mu) > 0\} \cap \{\ell : w^{s'}(\mu) > 0\} = \emptyset$ holds for any pair $w^s(\mu)$ and $w^{s'}(\mu)$. Consequently, even when sampling is carried out at each time t according to a randomly selected $w^{s_0}(\hat{\mu}(t))$, the final sampling proportion lies in a linear combination of the $w^s(\mu)$. This leads $Z_{s_0}(t) = \sum_{s \in \mathcal{C}(s_0)} Z_s(t) = t/d_{s_0}(\mu)$ for sufficiently large t and thus achieves asymptotic optimality.

G Additional Experiments

To validate the robustness of the proposed method for varying θ , we performed experiments with $V_{s_0}(\mu) = 0.6$ and $\theta = 0.5, 0.52, \dots, 0.58$. We randomly generated 500 trees for each θ , the same as Section 6, where each node has a branching factor randomly chosen from $\{2, 3, 4, 5\}$, and each node has children with probability 0.5 up to a maximum depth of 5. We set nodes at even depths as MAX nodes and those at odd depths as MIN nodes. Since we only evaluated tracking algorithms, we did not constrain the value of the second-best depth-1 node, which is for BAI algorithms.

For each method and each value of θ , we ran 10,000 times (20 iteration $\times$ 500 trees) and measured the average, 25%, and 75% percentile of the stopping time with $\delta = 10^{-2}$. Figure 2 shows the ratio of the stopping time to its asymptotic average lower bound $\mathbb{E}_{\mu \in \text{test cases}} [\log(1/\delta)/d_{s_0}(\mu)]$ averaged over 10,000 iterations. The shaded region enclosed by the dotted lines represent a 25% and 75% percentile of the stopping time. Table 1 shows the average stopping time of each method and the average asymptotic lower bound.

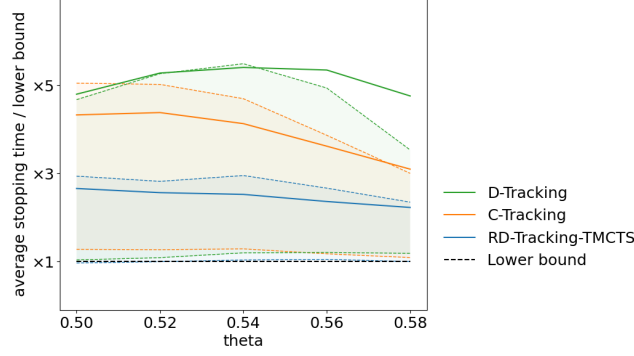


Figure 2: Ratio of the average stopping time to the lower bound for different values of θ . The shaded region enclosed by the dotted lines represent a 25% and 75% percentile.

Table 1: The average stopping time over 10,00 iterations.

θ	(Lower bound)	RD-Tracking-TMCTS	C-Tracking	D-Tracking
0.50	287.39	762.59	1242.01	1376.99
0.52	437.65	1119.97	1914.65	2306.96
0.54	744.01	1874.77	3068.27	4016.22
0.56	1586.60	3741.82	5732.31	8471.60
0.58	5892.44	13100.69	18218.92	27997.38

As a result, RD-Tracking-TMCTS outperformed for all θ . Moreover, while the average stopping time itself changes with the intrinsic difficulty of each instance (Table 1), the ratio to the theoretical lower bound remains stable (or improves) across different values of θ (Figure 2).

H Good Action Identification Problem

We consider the problem of identifying a depth-1 node whose value is at least θ . We call this problem the good action identification (GAI) problem and answer this question affirmatively. The correct answers of GAI problem $a^{\text{GAI}}(\mu)$ are defined as

$$a^{\text{GAI}}(\mu) = \begin{cases} \text{any } c \in A_{s_0}(\mu) & (A_{s_0}(\mu) \neq \emptyset) \\ \text{"no good action"} & (A_{s_0}(\mu) = \emptyset). \end{cases}$$

For GAI problem, $\liminf_{\delta \rightarrow 0} \mathbb{E}_{\mu}[\tau_{\delta}] / \log(1/\delta)$ is known to be lower bounded Degenne and Koolen [2019] by

$$\left(\max_{c \in A_{s_0}(\mu)} \max_{w \in \Delta} \inf_{\lambda \in \text{Alt}^{\text{GAI}}(c)} \sum_{\ell \in \mathcal{L}(\mathcal{T})} w_{\ell} d(\mu_{\ell}, \lambda_{\ell}) \right)^{-1}, \quad (24)$$

where

$$\text{Alt}^{\text{GAI}}(c) = \{\lambda \in X^{|\mathcal{L}(\mathcal{T})|} \mid c \notin A_{s_0}(\lambda)\}.$$

We can show that (24) is lower bounded by $d_{s_0}(\mu)^{-1}$ as the decision version. An algorithm for GAI problem is constructed by modifying the GLR statistic $Z_{s_0}(t)$ in the stopping rule (line 5 of Algorithm 1) as

$$Z_{s_0}^{\text{GAI}}(t) = \begin{cases} \max_{c \in \mathcal{C}(s_0)} Z_c(t) & (V_{s_0}(\hat{\mu}(t)) \geq \theta) \\ Z_{s_0}(t) & (V_{s_0}(\hat{\mu}(t)) < \theta). \end{cases}$$

and the output $a_{s_0}(\hat{\mu}(t))$ (line 6 of Algorithm 1) as

$$\hat{a}^{\text{GAI}}(t) = \begin{cases} \arg \max_{c \in \mathcal{C}(s_0)} Z_c^{\text{GAI}}(t) & (A_{s_0}(\hat{\mu}(t)) \neq \emptyset) \\ \text{"no good action"} & (\text{otherwise}). \end{cases}$$

888 Then, the modified algorithm satisfies

$$\mathbb{P}[\hat{a}^{\text{GAI}}(\tau_\delta) \text{ is a correct answer}] \geq 1 - \delta.$$

889 Since $Z_{s_0}^{\text{GAI}}(t) \leq Z_{s_0}(t)$, the empirical stopping time becomes larger than that of the thresholding
 890 problem. However, we can nevertheless derive the same upper bound on the asymptotic sample
 891 complexity.

892 **Theorem H.1.** *The algorithm obtained by replacing the stopping condition of Algorithm 1 (line 5)*
 893 *with $Z_{s_0}^{\text{GAI}}(t) \geq \beta(t, \delta)$ satisfies*

$$\limsup_{\delta \rightarrow +0} \frac{\mathbb{E}[\tau_\delta]}{\log(1/\delta)} \leq \frac{1}{d_{s_0}(\boldsymbol{\mu})}.$$

894 This result follows from the fact that, under sufficient convergence of the empirical means, the GLR
 895 statistics $Z_{s_0}(t)$ and $Z_{s_0}^{\text{GAI}}(t)$ grow at the same rate as functions of t . Since their growth behaviors
 896 are identical and the same threshold $\beta(t, \delta)$ is used, the resulting stopping times share the same
 897 asymptotic behavior.

898 H.1 δ -correctness

899 Here, we state the δ -correctness of the algorithm for the good action identification problem.

900 **Theorem H.2.** *The algorithm obtained by replacing the stopping rule of Algorithm 1 (line 5) with*
 901 *$Z_{s_0}^{\text{GAI}}(t) \geq \beta(t, \delta)$ satisfies*

$$\mathbb{P}(\tau_\delta < \infty, \hat{a}^{\text{GAI}}(\tau_\delta) \text{ is not a correct answer}) \leq \delta$$

902 *with the sequence of thresholds $(\beta(t, \delta))_{t \in \mathbb{N}}$ defined by Eq. (8).*

903 *Proof.* We define the set of mean vectors of reward distributions for which the answer $\hat{a}^{\text{GAI}}(t)$ is
 904 incorrect as

$$\text{Alt}^{\text{GAI}}(t) = \begin{cases} \{\boldsymbol{\lambda} = (\lambda_\ell)_{\ell \in \mathcal{L}(\mathcal{T})} \in X^{|\mathcal{L}(\mathcal{T})|} \mid V_{\hat{a}^{\text{GAI}}(t)}(\boldsymbol{\lambda}) < \theta\} & (\hat{a}^{\text{GAI}}(t) \neq \text{"no good action"}), \\ \{\boldsymbol{\lambda} = (\lambda_\ell)_{\ell \in \mathcal{L}(\mathcal{T})} \in X^{|\mathcal{L}(\mathcal{T})|} \mid V_{s_0}(\boldsymbol{\lambda}) \geq \theta\} & (\hat{a}^{\text{GAI}}(t) = \text{"no good action"}). \end{cases}$$

905 Here, the GLR statistics become

$$Z_{s_0}^{\text{GAI}}(t) = \log \frac{\sup_{\boldsymbol{\lambda} \in X^{|\mathcal{L}(\mathcal{T})|}} \Lambda(\boldsymbol{\lambda} \mid R_{s_0}(t))}{\sup_{\boldsymbol{\lambda} \in \text{Alt}^{\text{GAI}}(t)} \Lambda(\boldsymbol{\lambda} \mid R_{s_0}(t))}.$$

906 As in Corollary 4.2, by following the argument of Proposition 15 in Kaufmann and Koolen [2021],
 907 we can verify the δ -correctness of the stopping rule using $Z_{s_0}^{\text{GAI}}(t)$ of this form.

908 We next show that $Z_{s_0}^{\text{GAI}}(t)$ can be expressed as Eq. (H). First, we consider the case with $a_{s_0}(\hat{\boldsymbol{\mu}}(t)) =$
 909 ‘lose’. This implies $\hat{a}^{\text{GAI}}(t) = \text{"no good action"}$, and we obtain $\text{Alt}(\hat{\boldsymbol{\mu}}(t)) = \text{Alt}^{\text{GAI}}(t)$ and

$$\begin{aligned} Z_{s_0}^{\text{GAI}}(t) &= \log \frac{\sup_{\boldsymbol{\lambda} \in X^{|\mathcal{L}(\mathcal{T})|}} \Lambda(\boldsymbol{\lambda} \mid R_{s_0}(t))}{\sup_{\boldsymbol{\lambda} \in \text{Alt}^{\text{GAI}}(t)} \Lambda(\boldsymbol{\lambda} \mid R_{s_0}(t))} \\ &= \log \frac{\sup_{\boldsymbol{\lambda} \in X^{|\mathcal{L}(\mathcal{T})|}} \Lambda(\boldsymbol{\lambda} \mid R_{s_0}(t))}{\sup_{\boldsymbol{\lambda} \in \text{Alt}(\hat{\boldsymbol{\mu}}(t))} \Lambda(\boldsymbol{\lambda} \mid R_{s_0}(t))} = Z_{s_0}(t). \end{aligned}$$

910 Next, we consider the case with $a_{s_0}(\hat{\boldsymbol{\mu}}(t)) = \text{"win"}$. This implies $\hat{a}^{\text{GAI}}(t) \neq \text{"no good action"}$ and

$$\begin{aligned} \text{Alt}^{\text{GAI}}(t) &= \{\boldsymbol{\lambda} = (\lambda_\ell)_{\ell \in \mathcal{L}(\mathcal{T})} \in X^{|\mathcal{L}(\mathcal{T})|} \mid V_{a^{\text{GAI}}(t)}(\boldsymbol{\lambda}) < \theta\} \\ &= \{\boldsymbol{\lambda} = (\lambda_\ell)_{\ell \in \mathcal{L}(\mathcal{T})} \in X^{|\mathcal{L}(\mathcal{T})|} \mid (\lambda_{\ell'})_{\ell' \in \mathcal{D}(a^{\text{GAI}}(t))} \in \text{Alt}_{a^{\text{GAI}}(t)}(\hat{\boldsymbol{\mu}}(t))\}. \end{aligned}$$

911 From the definition of the likelihood, $\Lambda(\boldsymbol{\lambda} \mid R_{s_0}(t)) = \prod_{c \in \mathcal{C}(s_0)} \Lambda((\lambda_\ell)_{\ell \in \mathcal{C}(c)} \mid R_c(t))$ holds. Hence,
 912 we obtain

$$\begin{aligned}
 Z_{s_0}^{\text{GAI}}(t) &= \log \frac{\sup_{\boldsymbol{\lambda} \in X^{|\mathcal{L}(\mathcal{T})|}} \Lambda(\boldsymbol{\lambda} \mid R_{s_0}(t))}{\sup_{\boldsymbol{\lambda} \in \text{Alt}^{\text{GAI}}(t)} \Lambda(\boldsymbol{\lambda} \mid R_{s_0}(t))} \\
 &= \log \frac{\sup_{\boldsymbol{\lambda} \in X^{|\mathcal{L}(\mathcal{T})|}} \prod_{c \in \mathcal{C}(s_0)} \Lambda((\lambda_\ell)_{\ell \in \mathcal{C}(c)} \mid R_c(t))}{\sup_{\boldsymbol{\lambda} \in \text{Alt}^{\text{GAI}}(t)} \prod_{c \in \mathcal{C}(s_0)} \Lambda((\lambda_\ell)_{\ell \in \mathcal{C}(c)} \mid R_c(t))} \\
 &= \log \left(\frac{\prod_{c \in \mathcal{C}(s_0)} \sup_{(\lambda_\ell)_{\ell \in \mathcal{D}(c)} \in X^{|\mathcal{D}(c)|}} \Lambda((\lambda_\ell)_{\ell \in \mathcal{C}(c)} \mid R_c(t))}{\prod_{c \in \mathcal{C}(s_0) \setminus \{\hat{a}^{\text{GAI}}(t)\}} \sup_{(\lambda_\ell)_{\ell \in \mathcal{D}(c)} \in X^{|\mathcal{D}(c)|}} \Lambda((\lambda_\ell)_{\ell \in \mathcal{C}(c)} \mid R_c(t))} \right. \\
 &\quad \left. \times \frac{1}{\sup_{(\lambda_\ell)_{\ell \in \mathcal{D}(\hat{a}^{\text{GAI}}(t))} \in \text{Alt}_{\hat{a}^{\text{GAI}}(t)}(\hat{\boldsymbol{\mu}}(t))} \Lambda((\lambda_\ell)_{\ell \in \mathcal{C}(\hat{a}^{\text{GAI}}(t))} \mid R_{\hat{a}^{\text{GAI}}(t)}(t))} \right) \\
 &= \log \frac{\sup_{(\lambda_\ell)_{\ell \in \mathcal{D}(\hat{a}^{\text{GAI}}(t))} \in X^{|\mathcal{D}(\hat{a}^{\text{GAI}}(t))|}} \Lambda((\lambda_\ell)_{\ell \in \mathcal{C}(\hat{a}^{\text{GAI}}(t))} \mid R_{\hat{a}^{\text{GAI}}(t)}(t))}{\sup_{(\lambda_\ell)_{\ell \in \mathcal{D}(\hat{a}^{\text{GAI}}(t))} \in \text{Alt}_{\hat{a}^{\text{GAI}}(t)}(\hat{\boldsymbol{\mu}}(t))} \Lambda((\lambda_\ell)_{\ell \in \mathcal{C}(\hat{a}^{\text{GAI}}(t))} \mid R_{\hat{a}^{\text{GAI}}(t)}(t))} \\
 &= Z_{\hat{a}^{\text{GAI}}(t)}(t) \\
 &= \max_{c \in \mathcal{C}(s_0)} Z_c(t). \quad \square
 \end{aligned}$$

913 H.2 Proof of Theorem H.1

914 *Proof.* First, we consider the case with $a_{s_0}(\hat{\boldsymbol{\mu}}(t)) = \text{'lose'}$. From Eq. (H), we have $Z_{s_0}^{\text{GAI}}(t) =$
 915 $Z_{s_0}(t)$. Therefore, by following the same steps as in Theorem 5.4, we can verify that

$$Z_{s_0}^{\text{GAI}}(t) \geq td_{s_0}^\epsilon(\boldsymbol{\mu})$$

916 and

$$\limsup_{\delta \rightarrow +0} \frac{\mathbb{E}[\tau_\delta]}{\log(1/\delta)} \leq \frac{1}{d_{s_0}(\boldsymbol{\mu})}$$

917 are guaranteed to hold.

918 Next, we consider the case with $a_{s_0}(\hat{\boldsymbol{\mu}}(t)) = \text{'win'}$. From Eq. (H), we have $Z_{s_0}^{\text{GAI}}(t) =$
 919 $\max_{c \in \mathcal{C}(s)} Z_c(t)$. This implies that

$$\begin{aligned}
 Z_{s_0}^{\text{GAI}} &= \max_{c \in \mathcal{C}(s)} Z_c(t) \\
 &\geq \max_{c \in \mathcal{C}(s)} td_c^\epsilon(\boldsymbol{\mu}),
 \end{aligned}$$

920 where

$$d_c^\epsilon(\boldsymbol{\mu}) := \inf_{\boldsymbol{w}': \|\boldsymbol{w}' - \boldsymbol{w}^{s_0}(\boldsymbol{\mu})\| < 2(|\mathcal{L}(\mathcal{T})| - 1)\epsilon} \inf_{\boldsymbol{\mu}' \in \mathcal{I}_\epsilon} \inf_{\boldsymbol{\lambda} \in \text{Alt}(\hat{\boldsymbol{\mu}}(t))} \sum_{\ell \in \mathcal{D}(c)} \frac{N_\ell(t)}{t} d(\hat{\boldsymbol{\mu}}_\ell(t), \lambda_\ell).$$

921 By following the same steps as in Theorem 5.4, we obtain

$$\limsup_{\delta \rightarrow +0} \frac{\mathbb{E}_{\boldsymbol{\mu}}[\tau_\delta]}{\log(1/\delta)} \leq \frac{1 + \eta}{\max_{c \in \mathcal{C}(s)} d_c^\epsilon(\boldsymbol{\mu})}$$

922 for every $\eta > 0$ and $\epsilon > 0$.

923 Since $\lim_{\epsilon \rightarrow +0} d_c^\epsilon(\boldsymbol{\mu}) = d_c(\boldsymbol{\mu})$ and $d_{s_0}(\boldsymbol{\mu}) = \max_{c \in \mathcal{C}(s)} d_c(\boldsymbol{\mu})$, we conclude

$$\limsup_{\delta \rightarrow +0} \frac{\mathbb{E}[\tau_\delta]}{\log(1/\delta)} \leq \frac{1}{d_{s_0}(\boldsymbol{\mu})}. \quad \square$$

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